Universidade do Minho
Escola de Engenharia

Luís Paulo Peixoto dos Santos

Curriculum Vitæ

Habilitation in Informatics

January 2026

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Declaração de Integridade

Declaro ter atuado com integridade na elaboração do presente trabalho académico e confirmo que não recorri à prática de plágio nem a qualquer forma de utilização indevida ou falsificação de informações ou resultados em nenhuma das etapas conducente à sua elaboração.
Mais declaro que conheço e que respeitei o Código de Conduta Ética da Universidade do Minho.

Universidade do Minho, Braga, January 2026

Luís Paulo Peixoto dos Santos

Este texto constitui o *curriculum vitae* submetido à Universidade do Minho para preenchimento parcial dos requisitos para realização de Provas de Agregação no ramo do conhecimento de Informática nos termos do número 2 do artigo 8º do DL 239/2007, de 19 de Junho, cuja alínea c) menciona *"Currículo, com indicação do percurso profissional, das obras e dos trabalhos efectuados e das actividades científicas, tecnológicas e pedagógicas desenvolvidas, incluindo as suas actividades de investigação presentes e projectos e programas futuros;"*.

Contents

1 Identification

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- **Name:** Luís Paulo Peixoto dos Santos
- **Date of birth:** 29 November 1967
- **Place of birth:** Lisbon, Portugal
- **Nationality:** Portuguese
- **Citizen Card:** 7726535
- **Telephone/Fax:** 235 60 4439 / 253 60 4471
- **Web:** http://gec.di.uminho.pt/psantos
- **Email:** psantos@di.uminho.pt

1.1 Academic Qualifications

[Back to Identification]

2001 – PhD in Computer Science, University of Minho

1994 – Pedagogical Aptitude and Scientific Capacity Examination, University of Minho

1990 – Bachelor's degree in Systems and Computer Engineering, University of Minho

1.2 Professional Status and Career

[Back to Identification]

Place of Employment

Department of Informatics, School of Engineering

University of Minho

Gualtar Campus, 4710-054 Braga

Portugal

2024 – ... Associate Professor

2001 – 2024 Assistant Professor

1994 – 2001 Assistant Lecturer

1990 – 1994 Trainee Assistant Lecturer

1.3 Additional Affiliations

[Back to Identification]

2019 – ... *Cooperation Associate*, Iberian International Nanotechnology Laboratory (INL), *Quantum Linear Optics and Computation* group, Braga, Portugal

2022 – ... Senior Researcher, HASLAB – High Assurance Software Laboratory, INESC TEC

2016 – 2022 Senior Researcher, CSIG – Centre for Information Systems and Graphics, INESC TEC

1.4 Identifiers in Indexing Services

[Back to Identification]

Indexing Service	**Identifier**
ORCID ID	0000-0003-4466-1129
Scopus Author ID	36640396700
Researcher ID	A-1897-2009
Ciência ID	4B18-3424-EBF0
Google Scholar ID	0OyFWZ0AAAAJ
DBLP	16/2461

Identifiers in Scientific Publication Indexing Services

2 *Narrative Curriculum Vitæ*

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The following narrative curriculum presents the candidate's professional trajectory in a more discursive manner, complementing and contextualising some of the information conveyed in list form in the subsequent chapters. The main objective is to highlight and frame choices, outcomes and constraints, facilitating a qualitative analysis as opposed to a purely quantitative assessment.

2.1 Professional trajectory

[Back to the *Narrative Curriculum Vitæ*]

My professional and scientific career has unfolded entirely within the orbit of the Department of Informatics of the University of Minho. After completing the degree in Systems and Informatics Engineering in 1990, I began my academic career in this Department, with only a brief interruption to fulfil military service in the Portuguese Navy. In 1994 I completed the Pedagogical Aptitude and Scientific Capacity examinations, and in 2001 I was awarded the PhD degree in Informatics, both addressing topics in the field of Parallel Computing.

My scientific and pedagogical activity subsequently evolved across three areas: Parallel Computing (1990–2003), Computer Graphics (2003–2017), and Quantum Computing (2018–present). Rather than representing changes of field, these transitions correspond to a broadening of areas of interest. My more recent work frequently integrates challenges, results and methods associated with previous research areas, as clearly illustrated by items 10 (Quantum Computing + Graphics) and 24 (Computer Graphics + Parallel Computing) in Section 3.1. The motivation for reorienting my scientific interests has always been a combination of interaction with researchers who stimulated my curiosity and an attraction to the elegance of the mathematical models underpinning these areas of knowledge.

Aware that the quality of scientific research is strongly dependent on interaction with other researchers and alternative ways of thinking and working, I prioritised research stays at centres abroad. I spent several periods of between two and six months at other universities, namely the University of Bristol, UK (item 186), Université de Rennes, France (items 183 to 185), the University of Warwick, UK (item 182), and the

Texas Advanced Computing Center, Austin, TX, USA (items 179 and 180). The stays in Bristol, Rennes and Warwick were fundamental to the production of my main results in the area of Computer Graphics (for example, items 13, 20, 21 and 25). The stay in Austin, TX, USA enabled the production of results at the intersection of Parallel, Graphics and Quantum Computing (item 10).

Investment in Quantum Computing has begun to yield results more recently, in particular due to the excellent work of the three PhD students whom I have the pleasure of supervising (see items 241, 242 and 245). Of note are the eight articles published in first-quartile (Q1) scientific journals between 2023 and 2025 (items 4 to 11), as well as the large number of PhD examination committees in which I have participated in recent months (see Section 5.3). I also note two manuscripts currently under review for publication, available on arXiv (items 3 and 1).

My teaching activity has closely accompanied my scientific work. I have been responsible for the course unit “Computer Architecture”, in the second year of the Bachelor’s degree in Informatics Engineering, almost uninterruptedly since 2001 (items 190, 191, 192 and 194). I have also been responsible, since 2003, for a course unit addressing Image Synthesis and Global Illumination, offered in the first year of the Master’s degree in Informatics Engineering (items 198 to 201). Furthermore, since 2021 I have been responsible for the course unit “Quantum Data Science”, in the first year of the Master’s degree in Physical Engineering – specialisation in Information Physics (see item 197 and the corresponding Course Unit Report included in these proceedings).

2.2 Contributions to science, teaching and the institution

[Back to the *Narrative Curriculum Vitæ*]

Contributions to science and knowledge
Contributions to teaching
Institutional capacity building

Contributions to science, teaching and the institution — Quick access

Contributions to science and knowledge

[Back to Contributions]

Among the various results published over the years, I wish to highlight five contributions that I consider to be of particular relevance.

In accordance with point 3.d) of regulation VRT-ECF-04/2024 (*“Application for aggregation, habilitation or* *specialist examinations"*), I provide in Table 1 the handles of these open-access publications deposited in RepositóriUM, as an alternative to submitting them as additional documents:

"Reducing the resources required by ADAPT-VQE using coupled exchange operators and improved subroutines"	https://hdl.handle.net/1822/97017
"Trainability issues in quantum policy gradients"	https://hdl.handle.net/1822/97018
"Ensemble Metropolis Light Transport"	https://hdl.handle.net/1822/78131
"A Spherical Gaussian Framework for Bayesian Monte Carlo Rendering of Glossy Surfaces"	https://hdl.handle.net/1822/25139
"A Local Model of Eye Adaptation for High Dynamic Range Images"	https://hdl.handle.net/1822/17831

Table 1: Handles of these publications in RepositóriUM

- Mafalda Ramôa, Panagiotis G. Anastasiou, Luís Paulo Santos, Nicholas J. Mayhall, Edwin Barnes and Sophia E. Economou; **"Reducing the resources required by ADAPT-VQE using coupled exchange operators and improved subroutines"**; npj Quantum Information, vol. 11 (86), May, **2025**

  doi 10.1038/s41534-025-01039-4

  RepositóriUM https://hdl.handle.net/1822/97017

This article represents a highly significant milestone in the trajectory pursued by this group in the field of Quantum Computing. Published in a scientific journal of high prestige and impact (npj Quantum Information, a Nature partner journal, with a five-year impact factor of 9.3), it has the potential for wide visibility and for decisively influencing practice in the area of quantum molecular simulation.

The article proposes a new set of operators for adaptive quantum algorithms for eigenvalue estimation using the variational principle. This operator pool reduces quantum computational resources by approximately 90% relative to the best previously known operator set, and reduces measurement costs by five orders of magnitude compared to the best non-adaptive algorithms.

This contribution raises the expectation that the algorithm may be used in the near term to demonstrate quantum advantage, that is, to enable the simulation of molecules that are not simulable using classical resources alone. An experiment in this direction is currently being prepared in collaboration with IBM, with results expected in early 2026. Should these results prove to be statistically significant, this would constitute a high-visibility event with the potential to increase acceptance of quantum computing as a genuinely transformative technology.

- André Sequeira, Luís Paulo Santos, Luís Soares Barbosa; **"Trainability issues in quantum policy gradients"**; Machine Learning: Science and Technology, IOP Science, July, **2024**

  doi 10.1088/2632-2153/ad6830

  RepositóriUM https://hdl.handle.net/1822/97018

This article bears witness to the potential of the work that has been developed in the field of Quantum Computing, namely in Quantum Machine Learning (QML) and, in particular, in quantum Reinforcement Learning (RL).
QML has attracted considerable interest and generated high expectations with the development of variational quantum algorithms. These expectations have been increasingly overshadowed by results demonstrating the difficulty (and, in some cases, the impossibility) of efficiently training such models due to exponentially vanishing gradients. This article innovates by studying the trainability of these circuits in the context of quantum RL, demonstrating that while challenges exist, there is a trainable window for a polynomial number of actions and a continuous mapping of quantum states to those actions.

- Thomas Bashford-Rogers, Luís Paulo Santos, Demetris Marnerides, Kurt Debattista; **"Ensemble Metropolis Light Transport"**; ACM Transactions on Graphics, Volume 41(1), February, **2022**

  Invited presentation at SIGGRAPH 2022

  doi 10.1145/3472294

  RepositóriUM https://hdl.handle.net/1822/78131

This article stands out for having been published in the highest-impact scientific journal in the field of Computer Graphics and for having been presented, by invitation, at the most prestigious forum in the field: ACM SIGGRAPH. This publication also testifies to the continuity of my work in Computer Graphics, despite a reorientation of interests towards Quantum Computing.
The article presents an innovative image synthesis algorithm based on ensembles of Markov Chain Monte Carlo paths. The novelty lies in the use of global information extracted from the ensemble of paths to locally guide each individual path. Leveraging this information results in an improved convergence rate relative to the best algorithms known at the time. The multiplicity of paths allows for efficient parallelisation of the algorithm.

- Marques, Ricardo; Bouville, Christian; Ribardiére, Mickael; Santos, L. P.; Bouatouch, Kadi; **"A Spherical Gaussian Framework for Bayesian Monte Carlo Rendering of Glossy Surfaces"**; IEEE Transactions on Visualization and Computer Graphics, Volume 19(10), October, **2013**

  doi 10.1109/TVCG.2013.79

  RepositóriUM https://hdl.handle.net/1822/25139

Co-authors Christian Bouville and Kadi Bouatouch introduced the use of Bayesian Monte Carlo integration in global illumination algorithms, addressing only diffuse reflections. This article extends the original work to support materials with directional (though not specular) reflections. Bayesian Monte Carlo (BMC) integration originates in numerical integration and had previously been shown to converge more rapidly than classical Monte Carlo techniques due to more effective use of prior knowledge and information carried by the sample set. Although BMC is not yet practical for widespread use, this article broadens the range of approaches available for estimating the light transport equation, subject to resolving challenges such as hyperparameter determination.

- Ledda, Patrick; Santos, Luís Paulo; Chalmers, Alan; **"A Local Model of Eye Adaptation for High Dynamic Range Images"**; Afrigraph 2004, **2004**

  doi 10.1145/1029949.1029978

  RepositóriUM https://hdl.handle.net/1822/17831/

This article dates from an early phase of my involvement in Computer Graphics. It is pioneering in introducing a local model of luminance adaptation of the Human Visual System (HVS) in the context of mapping high dynamic range images to lower dynamic ranges. This model is imported from the field of psychophysics into Computer Graphics. Several algorithms based on perceptual (local and global) HVS models were subsequently proposed during this period, frequently citing this article. Modelling HVS adaptation did not enjoy the same level of popularity, but this does not diminish the innovative character of the work. At present, it has accumulated 106 citations in Scopus and 176 in Google Scholar, despite having been published in a relatively little-known conference with only three editions.

Contributions to teaching

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With regard to formal teaching, three contributions deserve particular emphasis:

1. In 2003, the University of Minho launched the first Master's degree in Computer Graphics and Virtual Environments in the national context. I was part of the team that created this educational project, its Steering Committee and its teaching staff. Since that date, uninterruptedly to the present, the University of Minho, under my responsibility and instruction, has maintained an advanced second-cycle training offer in physical-based Global Illumination (items 198 to 201). This programme covers ray tracing, global illumination algorithms, Monte Carlo and Metropolis numerical integration, as well as a formal approach to the numerical solution of the light transport integral. It is the most relevant training offer in this area in Portugal;

2. From October to December 2004 I taught Algorithms and Data Structures at the University of Timor Lorosae in Dili, East Timor (item 195). This was an unforgettable personal experience that altered my perspective on the world, humanity and society. The independence referendum had taken place five years earlier, and the declaration of independence two years earlier. The country lacked basic infrastructure, including buildings, road networks, sanitation, water distribution and electricity. Even so, the University of Timor Lorosae, like many other institutions, operated under the circumstances possible, with the support of the Foundation of Portuguese Universities. I have since lost contact with the students from that period, but I believe I contributed in some way to the empowerment of the approximately 50 young people whom I had the privilege to accompany;

3. In 2021 I began teaching the course unit "Quantum Data Science" within the Master's degree in Physical Engineering, specialisation in Information Physics (item 197). This is a pioneering training offer in Portugal and in many other regions, addressing content and skills in Quantum Machine Learning. This course unit is the subject of the Course Unit Report included in these proceedings.

Institutional capacity building

[Back to Contributions]

From 2012 to 2017 I was part of a working group whose objective—successfully achieved—was the installation of a Special Unit on Electronic Governance of the United Nations University on a campus of the University of Minho (item 406). This initiative was formalised in 2016 as the Special Project of the University of Minho for Electronic Governance (UMinho-EGOV), with the UNU–EGOV Unit having been created in 2014 on the Couros Campus, Guimarães, and remaining in operation to this date.

This was a project quite different from my previous experience, involving stakeholders from very diverse spheres of activity, including academics, public sector technicians, and municipal and governmental officials. As a project with a strong political decision-making component, I did not always feel that my profile and experience were the most suitable. Nevertheless, it constituted an important learning experience regarding the functioning of public affairs and the importance of teamwork.

Within the scope of this initiative, I chaired the organisation of the ICEGOV 2014 conference (item 171), participated in two international cooperation projects (items 157 and 158), and published two conference papers in the area of Electronic Governance (items 45 and 46).

2.3 Plans for the near future

[Back to the *Narrative Curriculum Vitæ*]

At the time of writing this document, assuming no unforeseen circumstances, I have nine years of active professional life remaining until retirement. This is still a significant period that allows for the pursuit of several projects, though not for the embrace of entirely new fields of knowledge. I view this period as a phase of consolidation of initiatives already underway, seeking to equip the department and research centre with the human, financial and material resources necessary for younger colleagues to continue these lines of work, should they so wish.

There are four dimensions that I intend to prioritise in the near future:

- Research and teaching in Quantum Computing began in this department in 2018, as a result of the interest, dynamism and voluntary commitment of a small group of academics. Seven years on, considerable progress has been made, but medium-term sustainability is far from assured. Securing competitive funding is a fundamental requirement in order to expand and rejuvenate the body of academics and researchers in the area and to increase the visibility and impact of this group within the international scientific community.

- With the impending retirement of the majority of staff academics associated with the teaching of Computer Graphics and Quantum Computing, it is urgent to renew this teaching staff, at the risk of these areas disappearing from the University of Minho's educational offer.

- I intend to focus my research on the efficient classical simulation of quantum circuits. I am interested in solutions to this problem that establish a bridge between the domains of knowledge in which I have worked throughout my career. The simulation of quantum processes can be formulated as a sum over paths, leading naturally to probabilistic path integration methods, similar to those used in advanced image synthesis algorithms. Exploring these probabilistic algorithms (Monte Carlo and derivatives) in the context of quantum process simulation may enable the provision of more efficient tools to the scientific community. There are several challenges and differences relative to their use in Computer Graphics. It is not expected that a classical algorithm, probabilistic or otherwise, will allow efficient (non-exponential) simulation of arbitrary quantum circuits, as this would imply no separation between the two paradigms. The numerical sign problem may ultimately be a limiting factor in efficiency, since the addition of amplitudes from a set of paths (each with a non-zero complex amplitude) may result in a null contribution due to destructive interference. This approach is well suited to parallelisation at multiple levels (vectorisation, multithreading and multiple nodes), lends itself to modelling noise as a stochastic process, and does not exclude hybrid approaches (combining Schrödinger and Feynman simulations).

- I am interested in participating more frequently and more actively in science communication projects and initiatives, particularly in promoting scientific knowledge among civil society and graduate students, as well as knowledge about scientific knowledge and other forms of knowledge. In an era saturated with information sources of variable quality and pseudo-knowledge, it is imperative to foster critical thinking and awareness of the many fallacies associated with false, partial and incomplete information, as well as flawed inference processes.

3 Scientific Results

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This chapter characterizes the candidate's scientific activity. It is organized in the following sections:

- Publications;
- Community's Recognition;
- Projects' Coordination and Participation;
- Research Activities Coordination.

3.1 Publications

[Back to Scientific Results]

The table below presents the number of indexed papers and the number of citations, as published by the different indexing services on the 31st of August, 2025.

		Scopus		ISI Web of Science		Google Scholar	
Tipo	Tot.	Index	Cit.	Index	Cit.	Index	Cit.
Pre-prints	3	- -	- -	- -	- -	3	1
Journals	24	22	151	21	126	23	327
Books	1	1	7	- -	- -	1	10
Book chapters	2	- -	- -	- -	- -	2	2
Editorials	8	7	3	8	3	8	6
Int. Conferences	29	21	172	14	20	28	327
Nat. Conferences	6	1	1	1	0	6	18
Short papers	5	- -	- -	- -	- -	4	3
Posters	8	- -	- -	- -	- -	- -	- -
Total	86	52	334	45	149	75	694

Totals: indexing and citations

The table below presents the h-index, as published by the different indexing services on the 31st of August, 2025.

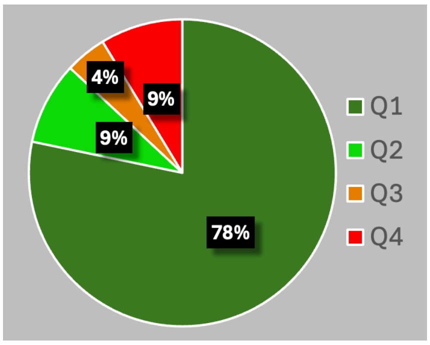
Indexing	***h-index***
Scopus	9
Web of Science	7
Google Scholar	14

h-index as published by different indexing services

The table below presents the number of journal articles per quartile, according to the Scimago Journal Ranking.

SJR Scimago Journal Ranking

Q1	Q2	Q3	Q4
18	2	1	2



Number of journal articles per quartile

Legend The listingss below include the following figures:

SJR – journal quartile as published by the Scimago Journal Rank;

Scopus – number of citations in Scopus (- - - if not indexed);

WEB OF SCIENCE – number of citations in Web of Science (- - - if not indexed);

– number of citations in Google Scholar (- - - if not indexed);

Pre-prints

[Back to Publications]

1. Alexandra Ramôa and Luís Paulo Santos and Akihito Soeda; **"Low Cost Bayesian Experimental Design for Quantum Frequency Estimation with Decoherence"**; Aug, **2025**

   Submitted to Quantum Science and Technology, IOP Science

   doi 10.48550/arXiv.2508.07120

   Scopus - - - WEB OF SCIENCE - - - 0

2. Thomas Bashford-Rogers, Luis Paulo Santos; **"Path Space Partitioning and Guided Image Sampling for MCMC"**; jan, **2025**

   doi 10.48550/arXiv.2501.06214

   Scopus - - - WEB OF SCIENCE - - - 0

3. Bruna Salgado and André Sequeira and Luis Paulo Santos; **"A hybrid classical-quantum approach to highly constrained Unit Commitment problems"**; Dec, **2024**

   Submitted to IEEE Transactions on Quantum Engineering

10.48550/arXiv.2412.11312

Scopus --- WEB OF SCIENCE --- 1

Journals

[Back to Publications]

4. Alexandra Ramôa, Luis Paulo Santos; **"Bayesian Quantum Amplitude Estimation"**; Quantum, vol. 9 (1856), August, **2025**

   10.22331/q-2025-09-11-1856 SJR Q1

   Scopus --- WEB OF SCIENCE --- 2

5. Mafalda Ramôa, Panagiotis G. Anastasiou, Luis Paulo Santos, Nicholas J. Mayhall, Edwin Barnes and Sophia E. Economou ; **"Reducing the resources required by ADAPT-VQE using coupled exchange operators and improved subroutines"**; npj Quantum Information, vol. 11 (86), May, **2025**

   10.1038/s41534-025-01039-4 SJR Q1

   Scopus 1 WEB OF SCIENCE 2 12

6. Mafalda Ramôa, Luís Paulo Santos; **"Reducing Measurement Costs by Recycling the Hessian in Adaptive Variational Quantum Algorithms"**; Quantum Science and Technology, IOP Science, vol. 10 (1), november, **2024**

   10.1088/2058-9565/ad904e SJR Q1

   Scopus 0 WEB OF SCIENCE 1 4

7. Rodrigo Coelho, André Sequeira, Luís Paulo Santos; **"VQC-Based Reinforcement Learning with Data Re-uploading: Performance and Trainability"**; Quantum Machine Intelligence, Springer, vol. 6 (53), August, **2024**

   10.1007/s42484-024-00190-z SJR Q1

   Scopus 5 WEB OF SCIENCE 3 14

8. André Sequeira, Luís Paulo Santos, Luís Soares Barbosa; **"Trainability issues in quantum policy gradients"**; Machine Learning: Science and Technology, vol. 5 (3), IOP Science, July, **2024**

   10.1088/2632-2153/ad6830 SJR Q1

   Scopus 1 WEB OF SCIENCE 0 3

9. André Sequeira, Luís Paulo Santos, Luís Soares Barbosa; **"On Quantum Natural Policy Gradients"**; IEEE Transactions on Quantum Engineering, vol. 5, June, **2024**

   doi 10.1109/TQE.2024.3418094 SJR Q1

   Scopus 1 WEB OF SCIENCE 0 3

10. Luís Paulo Santos, Thomas Bashford-Rogers, João Barbosa, Paul Navrátil; **"Towards Quantum Ray Tracing"**; IEEE Transactions on Visualization and Computer Graphics, April, **2024**

    doi 10.1109/TVCG.2024.3386103 SJR Q1

    Scopus 0 WEB OF SCIENCE 0 11

11. André Sequeira, Luís Paulo Santos, Luís Soares Barbosa; **"Policy gradients using variational quantum circuits"**; Quantum Machine Intelligence, Volume 5(18), April, **2023**

    doi 10.1007/s42484-023-00101-8 SJR Q1

    Scopus 12 WEB OF SCIENCE 9 30

12. Arnau Colom, Ricardo Marques, Luís Paulo Santos; **"Interactive VPL-based Global Illumination on the GPU using Fuzzy Clustering"**; Computers & Graphics, Volume 108, November, **2022**

    doi 10.1016/j.cag.2022.09.008 SJR Q2

    Scopus 2 WEB OF SCIENCE 1 2

13. Thomas Bashford-Rogers, Luís Paulo Santos, Demetris Marnerides, Kurt Debattista; **"Ensemble Metropolis Light Transport"**; ACM Transactions on Graphics, Volume 41(1), February, **2022**

    Invited presentation at SIGGRAPH'2022

    doi 10.1145/3472294 SJR Q1

    Scopus 5 WEB OF SCIENCE 4 10

14. André Sequeira, Luís Paulo Santos, Luís Soares Barbosa; **"Quantum Tree-Based Planning"**; IEEE Access, Volume 9, **2021**

    doi 10.1109/ACCESS.2021.3110652 SJR Q1

    Scopus 2 WEB OF SCIENCE 2 3

15. Gabriel Falcão, Filipe Cabeleira, Artur Mariano, Luís Paulo Santos; **"Heterogeneous implementation of a Voronoi cell-based SVP solver"**; IEEE Access, Volume 7, September, **2019**

    doi 10.1109/10.1109/ACCESS.2019.2939142 SJR Q1

    Scopus 3 WEB OF SCIENCE 3 8

16. Roberto Ribeiro, João Barbosa, Luís Paulo Santos; **"A Framework for Efficient Execution of Data Parallel Irregular Applications on Heterogeneous Systems"**; Parallel Processing Letters, Volume 25(2), June, **2015**

    doi 10.1109/10.1142/S0129626415500048 SJR Q3

    Scopus 2 WEB OF SCIENCE 2 5

17. Vânio Ferreira, Luís Paulo Santos, Markus Franzen, Omar O. Ghouati, Ricardo Simões; **"Improving FEM crash simulation accuracy through local thickness estimation based on CAD data"**; Elsevier Advances in Engineering Software, Volume 71, **2014**

    doi 10.1016/j.advengsoft.2014.02.003 SJR Q1

    Scopus 10 WEB OF SCIENCE 8 12

18. Luis Unzueta, Waldir Pimenta, Jon Goenetxea, Luís Paulo Santos, Fadi Dornaika; **"Efficient Generic Face Model Fitting to Images and Videos"**; Elsevier Image and Vision Computing, Volume 32(5), May, **2014**

    doi 10.1016/j.imavis.2014.02.006 SJR Q1

    Scopus 16 WEB OF SCIENCE 15 32

19. Nuno Silva, Luís Paulo Santos; **"Interactive High Fidelity Visualization of Complex Materials on the GPU"**; Elsevier Computers & Graphics, Volume 37(7), November, **2013**

    doi 10.1016/j.cag.2013.06.006 SJR Q1

    Scopus 2 WEB OF SCIENCE 2 2

20. R. Marques, C. Bouville, M. Ribardière, L.P. Santos, K. Bouatouch; **"Spherical Fibonacci Point Sets for Illumination Integrals"**; Computer Graphics Forum, Volume 32(8), November, **2013**

    doi 10.1111/cgf.12190 SJR Q1

    Scopus 55 WEB OF SCIENCE 50 84

21. Marques, Ricardo; Bouville, Christian; Ribardiére, Mickael; Santos LP, Bouatouch, Kadi; **"A Spherical Gaussian Framework for Bayesian Monte Carlo Rendering of Glossy Surfaces";** IEEE Transactions on Visualization and Computer Graphics, Volume 19(10), October, **2013**

    doi 10.1109/TVCG.2013.79 SJR Q1

    Scopus 13 WEB OF SCIENCE 16 21

22. Fernandes JM, van Hattum-Janssen N, Fonte V, Ribeiro AN, Santos LP, Sousa P; **"An integrated approach to develop professional and technical skills for informatics engineering stu-**

dents"; European Journal of Engineering Education, Volume 37(2), May, **2012**

10.1080/03043797.2012.666517 SJR Q1

Scopus 9 WEB OF SCIENCE 7 16

23. Ferreira, V. and Santos, L.P. and Franzen, M. and Ghouati, O.O. and Simoes, R.; **"Estimating Local Part Thickness in Midplane Meshes for Finite Element Analysis"**; International Journal of Mathematics and Computers In Simulation, Volume 5(1), **2011**

    https://www.naun.org/main/NAUN/mcs/19-822.pdf SJR Q4

    Scopus 3 WEB OF SCIENCE --- 3

24. Debattista, Kurt and Dubla, Piotr and Santos, Luís Paulo and Chalmers, Alan; **"Wait-Free Shared-Memory Irradiance Caching"**; IEEE Computer Graphics and Applications, Volume 31(5), **2011**

    10.1109/MCG.2010.80 SJR Q2

    Scopus 2 WEB OF SCIENCE 1 15

25. Debattista, Kurt and Dubla, Piotr and Banterle, Francesco and Santos, Luís Paulo and Chalmers, Alan; **"Instant Caching for Interactive Global Illumination"**; Computer Graphics Forum, Volume 28(8), December, **2009**

    10.1111/j.1467-8659.2009.01435.x SJR Q1

    Scopus 7 WEB OF SCIENCE 6 12

26. Chalmers, Alan; Debattista, Kurt; Mastoropoulou, Georgia; Santos, Luís Paulo; **"There-Reality: Selective Rendering in High Fidelity Virtual Environments";** International Journal of Virtual Reality, Volume 6(1), March, **2007**

    Link PDF SJR ---

    Scopus --- WEB OF SCIENCE --- 29

27. Santos, Luís Paulo; Chalmers, Alan; Proença, Alberto; **"A messages-density monitoring strategy for distributed-memory parallel systems"**; PROGRAMMING AND COMPUTER SOFTWARE, Volume 21(1), January, **1995**

    Link SJR Q4

    Scopus --- WEB OF SCIENCE --- 4

Books

[Back to Publications]

28. Ricardo Marques, Christian Bouville, Luís Paulo Santos, Kadi Bouatouch; **"Efficient Quadrature Rules for Illumination Integrals: From Quasi Monte Carlo to Bayesian Monte Carlo"**; Synthesis Lectures on Computer Graphics and Animation, Morgan and Claypool Publishers, eBook ISBN 978-3-031-79567-1; June, **2015**

    doi 10.1007/978-3-031-79567-1

    Scopus 7 WEB OF SCIENCE --- 10

Books' chapters

[Back to Publications]

29. André Mariano, Filipe Cabeleira, Luís Paulo Santos, Gabriel Falcão; **"Optimized Voronoi-based Algorithms for Parallel Shortest Vector Computation"**; Chapter 4 in "Cybersecurity & High-Performance Computing Environments: Integrated Innovations, Practices, and Applications"; Edited by Kuan-Ching Li, Nitin Sukhija, Elizabeth Bautista, Jean-Luc Gaudiot; CRC PRESS Taylor & Francis, ISBN 9781003155799, **2022**

    doi 10.1201/9781003155799-4

    Scopus --- WEB OF SCIENCE --- 0

30. Luis Unzueta, Waldir Pimenta, Jon Goenetxea, Luís Paulo Santos, Fadi Dornaika; **"Efficient Deformable 3D Face Model Fitting to Monocular Images"**; Chapter 8 in "Advances in Face Image Analysis: Theory and Applications" Fadi Dornaika (ed.); Bentham eBooks; eISBN: 978-1-68108-110-6, ISBN: 978-1-68108-111-3E; **2015**

    doi 10.2174/97816810811061160100l1

    Scopus --- WEB OF SCIENCE --- 2

Editorials

[Back to Publications]

This subsection lists editorials published as the guest editor of special sections in scientific journals.

31. Nuno Rodrigues, Daniel Mendes, Luís Paulo Santos and Kadi Bouatouch; **"Foreword to the special section on recent advances in graphics and interaction"**; Elsevier Computers & Graphics, volume 102, **2021**

    doi 10.1016/j.cag.2021.12.006 SJR Q2

    Scopus 0 WEB OF SCIENCE 0 0

32. Durikovic, Roman; Santos, Luís Paulo; **"Foreword to the Special Section on the Spring Conference on Computer Graphics 2015 (SCCG'2015)"**; Elsevier Computers & Graphics, volume 53, **2015**

doi 10.1016/j.cag.2015.05.030 SJR Q2

Scopus 0 WEB OF SCIENCE 0 0

33. Santos, Luís Paulo; Debattista, Kurt; **"Guest Editorial – High Dynamic Range Imaging"**; Elsevier Computers & Graphics, volume 37(7), **2013**

doi 10.1016/j.cag.2013.08.001 SJR Q2

Scopus --- WEB OF SCIENCE 1 1

34. Chalmers, Alan; Mudge, Mark; Santos, Luís Paulo; **"Special Section on Graphics for Cultural Heritage"**; Elsevier Computers & Graphics, volume 36(1), **2012**

doi 10.1016/j.cag.2011.11.006 SJR Q2

Scopus 1 WEB OF SCIENCE 0 1

35. Chalmers, Alan; Mudge, Mark; Santos, Luís Paulo; **"Special Section on Cultural Heritage"**; Elsevier Computers & Graphics, volume 35(4), **2011**

doi 10.1016/j.cag.2011.03.041 SJR Q2

Scopus 1 WEB OF SCIENCE 0 1

36. Debattista, Kurt; Proença, Alberto José; Santos, Luís Paulo; **"Preface and biographic notes for the special issue on graphics for serious games"**; Elsevier Computers & Graphics, volume 34(6), **2010**

doi 10.1016/j.cag.2010.09.016 SJR Q2

Scopus 0 WEB OF SCIENCE 0 0

37. Santos, Luís Paulo; Reiners, Dirk; Favre, Jean; **"Parallel Graphics and Visualization"**; Elsevier Computers & Graphics, volume 32(1), **2008**

doi 10.1016/j.cag.2007.12.004 SJR Q2

Scopus 1 WEB OF SCIENCE 2 3

38. Santos, Luís Paulo; Alan Heirich; Bruno Raffin; **"Parallel Graphics and Visualization"**; Parallel Computing, volume 33(6), **2007**

doi 10.1016/j.parco.2007.04.001 SJR Q2

Scopus 0 WEB OF SCIENCE 0 0

International Conferences

[Back to Publications]

39. André Sequeira, Luis Paulo Santos, Luís Soares Barbosa; **“Trainability issues in quantum policy gradients with softmax activations”**; IEEE Quantum Week (QCE 2024) in Workshop on Quantum Computing and Reinforcement Learning (QCRL-2024), **2024**

    QCE’ 2024 - QCRL-2024

    Scopus 0 WEB OF SCIENCE 0 0

40. João Barbosa, Luis Paulo Santos, Don Fussell and Paul Navratil; **“LOOM: Interweaving tightly coupled visualization and numeric simulation framework”**; Workshop on In Situ Infrastructures for Enabling Extreme-scale Analysis and Visualization (ISAV 2021), **2021**

    doi 10.1145/3490138.3490139

    Scopus 1 WEB OF SCIENCE - - - 2

41. André Sequeira, Luis Paulo Santos and Luis Soares Barbosa; **“Generalised Quantum Tree Search”**; Second International Workshop on Quantum Software Engineering (Q-SE 2021), **2021**

    doi 10.1109/Q-SE52541.2021.00015

    Scopus 0 WEB OF SCIENCE 0 1

42. Alves C., Santos L.P., Bashford-Rogers T.; **“A Quantum Algorithm for Ray Casting using an Orthographic Camera”**; 2019 Int. Conf. on Graphics and Interaction (ICGI), pp 56-63, IEEE Xplore, **2019**

    Prémio “Melhor Artigo”

    doi 10.1109/ICGI47575.2019.8955061

    Scopus 7 WEB OF SCIENCE 4 9

43. Ribeiro R., Santos L.P., Nóbrega J.M.; **“Runtime Heterogeneous Aware Power Adaptive Scheduling in OpenFOAM”**; 2018 Int. Conf. on High Performance Computing & Simulation (HPCS), IEEE Xplore, **2018**

    doi 10.1109/HPCS.2018.00070

    Scopus 0 WEB OF SCIENCE 0 0

44. Ribeiro R., Santos L.P., Nóbrega J.M.; **“nSharma: Numerical Simulation Heterogeneity Aware Runtime Manager for OpenFOAM”**; In: Shi Y. et al. (eds) Computational Science

– ICCS 2018. Lecture Notes in Computer Science, vol 10860. Springer, Cham, **2018**

doi 10.1007/978-3-319-93698-7_33

Scopus 1 WEB OF SCIENCE 0 2

45. Luís Paulo Santos, Luís Nuno Barbosa, Diogo Aires Bessa, Lúcia Pereira Martins, Luís Soares Barbosa; **"Communities of Practice as a tool to support the GCIO function"**; 11th International Conference on Theory and Practice of Electronic Governance (ICEGOV'2018), **2018**

doi 10.1145/3209415.3209507

Scopus 1 WEB OF SCIENCE 0 2

46. Luís Soares Barbosa, Luís Paulo Santos; **"Networks of Universities as a Tool for GCIO Education"**; International Conference on Electronic Government; Lecture Notes in Computer Science, 10428, pp. 117–127, **2017**

doi 10.1007/978-3-319-64677-0_10

Scopus 1 WEB OF SCIENCE 1 4

47. Artur Mariano, Ricardo Alves, Joao Barbosa, Luis Paulo Santos and Alberto Proença; **"A (ir)regularity-aware task scheduler for heterogeneous platforms"**; International Conference on High Performance Computing; **2012**

https://core.ac.uk/display/55626498

Scopus --- WEB OF SCIENCE --- 7

48. Alves, Albano; Rufino, José; Pina, António; Santos, Luís Paulo; **"clOpenCL – Supporting Distributed Heterogeneous Computing in HPC Clusters"**; Tenth International Workshop HeteroPar'2012 -Algorithms, Models and Tools for Parallel Computing on Heterogeneous Platforms; Lecture Notes in Computer Science 7640 (Euro-Par 2012 Workshops), Springer, **2012**

doi 10.1007/978-3-642-36949-0_14

Scopus 20 WEB OF SCIENCE 13 27

49. Silva , Nuno; Santos , Luís Paulo; Fussell , Donald;
"Real-Time Visualization of a Sparse Parametric Mixture Model for BTF Rendering"; International Symposium on Visual Computing 2012; Lecture Notes in Computer Science 7431 (ISVC 2012), Springer, **2012**

doi 10.1007/978-3-642-33179-4_68

Scopus 1 WEB OF SCIENCE 1 1

50. Pimenta, W., Santos, L.P.; **“A Comprehensive Taxonomy for Three-dimensional Displays”**; 20th WSCG International Conference on Computer Graphics, Visualization and Computer Vision **2012**

Scopus® 4 WEB OF SCIENCE 1 6

51. Carvalho, Elizabeth; Domingues, Hugo; Maçães, Gustavo; Santos, Luís Paulo; **“Augmented reality visualization and edition of cognitive workflow capturing”**; 1st Experiment@ International Conference **2011**

Scopus® --- WEB OF SCIENCE --- 2

52. Ferreira, Vânio and Santos, Luís Paulo and Simões, Ricardo and Franzen, Markus and Ghouati, Omar; **“Estimating Local thickness for Finite Element Analysis”**; 12th WSEAS International Conference on Mathematical and Computational Methods in Science and Engineering (MACMESE’2010), **2010**

Scopus® 1 WEB OF SCIENCE --- 1

53. Dubla, Piotr and Banterle, Francesco and Debattista, Kurt and Santos, Luís Paulo and Chalmers, Alan; **“Wait-Free Shared-Memory Irradiance Cache”**; Eurographics Symposium on Parallel Graphics and Visualization (EGPGV2009), **2009**

Scopus® --- WEB OF SCIENCE --- ---

54. Chalmers, Alan; Debattista, Kurt; Santos, Luís Paulo; **“Selective Rendering: Computing Only What You See”**; Graphite’2006 – Keynote Paper, **2006**

doi 10.1145/1174429.1174431

Scopus® 18 WEB OF SCIENCE --- 22

55. Oliveira, António and Santos, Luís Paulo and Proença, Alberto; “Refinement Criteria for High Fidelity Interactive Walkthroughs”; Graphite’2006, **2006**

doi 10.1145/1174429.1174505

Scopus® 0 WEB OF SCIENCE --- 0

56. Debattista, Kurt and Santos, Luís Paulo and Chalmers, Alan; **“Accelerating the Irradiance Cache through Parallel Component-Based Rendering”**; EGPGV’2006 - 6th Eurographics Symposium on Parallel Graphics Visualization, **2006**

doi 10.2312/EGPGV/EGPGV06/027-034

Scopus® --- WEB OF SCIENCE --- 20

57. Santos, Luís Paulo and Valentim, Sérgio and Fernandes, António Ramires; **“Parallel Progressive Precomputed Radiance Transfer”**; SCCG’2006 - Spring Conference on Computer Graphics, **2006**

doi 10.1145/2602161.2602165

Scopus 0 WEB OF SCIENCE --- 0

58. Debattista, K. and Sundstedt, V. and Santos, Luís Paulo and Chalmers, A.; **“Selective Component Based Rendering”**; Graphite’2005, **2005**

doi 10.1145/1101389.1101393

Scopus 17 WEB OF SCIENCE --- 26

59. Trystram, D. and Bender, M. and Schwiegelshohn, U. and Santos, Luís Paulo; **“Scheduling and Load Balancing”**; EuroPar’2005: Parallel Processing; Lecture Notes in Computer Science 3648; Springer-Verlag, **2005**

doi 10.1007/11549468_25

Scopus 0 WEB OF SCIENCE --- 0

60. Ledda, Patrick and Santos, Luis Paulo and Chalmers, Alan; **“A Local Model of Eye Adaptation for High Dynamic Range Images”**; Afrigraph’2004, **2004**

doi 10.1145/1029949.1029978

Scopus 106 WEB OF SCIENCE --- 176

61. Santos, Luis Paulo and Proença, Alberto; **“Scheduling Under Conditions of Uncertainty: a Bayesian Approach”**; EuroPar’2004: Parallel Processing; Lecture Notes in Computer Science 3149; Springer-Verlag, **2004**

doi 10.1007/978-3-540-27866-5_29

Scopus 3 WEB OF SCIENCE 2 8

62. Santos, Luis Paulo and Proença, Alberto; **“A Systematic Approach to Effective Scheduling in Distributed Systems”**; VECPAR’2002 - 5th Int. Meeting on High Performance Computing for Computational Science, **2002**

Scopus --- WEB OF SCIENCE --- 3

63. Santos, Luis Paulo and Proença, Alberto; **“A Bayesian RunTime Load Manager on a Shared Cluster”**; Scheduling and Load Balancing on Clusters (SLAB’2001), special session in IEEE International Symposium on Cluster Computing and the Grid (CCGrid’2001), **2001**

doi 10.1109/CCGRID.2001.923259

Scopus 0 WEB OF SCIENCE 0 4

64. Santos, Luis Paulo and Castro, Vítor and Proença, Alberto; **"Evaluation of the communication performance on a parallel processing system"**; Recent Advances In Parallel Virtual Machine and Message Passing Interface; Lecture Notes in Computer Science, vol. 1332, **1997**

    doi 10.1007/3-540-63697-8_68

    Scopus 0 WEB OF SCIENCE 0 4

65. Cunha, Alcino and Santos, Luís Paulo and Belo, Orlando; **"Enhancing load distribution strategies through simulation"**; 9th European Simulation Symposium, **1997**

    Scopus --- WEB OF SCIENCE 0 0

66. Santos, Luis Paulo and Chalmers, Alan and Proença, Alberto; **"A teaching lab on parallel processing"**; VECPAR'1993 - 1st Int. Meeting on Vector and Parallel Processing, **1993**

    Scopus --- WEB OF SCIENCE --- 0

67. Santos, Luis Paulo and Chalmers, Alan and Proença, Alberto; **"A data management strategy for increased parallel processing efficiency"**; VECPAR'1993 - 1st Int. Meeting on Vector and Parallel Processing, **1993**

    Scopus --- WEB OF SCIENCE --- 0

National Conferences

[Back to Publications]

68. Oliveira, André; Perdigão, César, Santos, Luís Paulo e Proença, Alberto; **"Exploring Heterogeneous Computing with Advanced Path Tracing Algorithms"**; Encontro Português de Computação Gráfica e Interacção, **2016**

    Prémio "Melhor Artigo"

    doi 10.1109/EPCGI.2016.7851185

    Scopus 1 WEB OF SCIENCE 0 0

69. Silva, Nuno e Santos, Luís Paulo; **"Visualização em Tempo Real de um Modelo Esparso de Mistura Paramétrica para síntese da BTF"**; 20° Encontro Português de Computação Gráfica, **2012 Prémio "Melhor Artigo"**

doi 10.2312/pt.20121103

Scopus --- WEB OF SCIENCE --- 0

70. Marques, Ricardo and Santos, Luís Paulo; **“Instant Global Illumination on the GPU using OptiX”**; Inforum, **2010**

https://core.ac.uk/display/55615652

Scopus --- WEB OF SCIENCE --- 2

71. Marques, Ricardo and Santos, Luís Paulo; **“GPU Ray Casting”**; 17° Encontro Português de Computação Gráfica, **2009**

Menção Honrosa

doi 10.2312/pt.20091215

Scopus --- WEB OF SCIENCE --- 12

72. Nunes, Miguel and Santos, Luís Paulo; **“Workload Distribution for Ray Tracing in Multi-Core Systems”**; 17° Encontro Português de Computação Gráfica, **2009**

doi 10.2312/pt.20091230

Scopus --- WEB OF SCIENCE --- 3

73. Santos, Luís Paulo and Sobral, João Luís; **“Inventário Automático de Sinais de Trânsito: um Sistema de Mapeamento Móvel”**; 15° Encontro Português de Computação Gráfica, **2007**

https://core.ac.uk/download/pdf/55619367.pdf

Scopus --- WEB OF SCIENCE --- 1

Short Papers

[Back to Publications]

74. Ramoa, Alexandra; Santos, Luis; **“Bayesian Amplitude Estimation”**; ReAQCT 2024: Conference on Recent Advances in Quantum Computing and Technology, **2024**

doi https://indico.wigner.hu/event/1588/contributions/4051/

Scopus --- WEB OF SCIENCE --- ---

75. Marques, Ricardo; Bouville, Christian; Santos, Luis; Bouatouch, Kadi; **“Two-Level Adaptive Sampling for Illumination Integrals using Bayesian Monte Carlo”**; EuroGraphics 2016: The 37th Annual Conference of the European Association for Computer Graphics (short paper), **2016**

https://dl.acm.org/doi/abs/10.5555/3059107.3059129

Scopus --- WEB OF SCIENCE --- 0

76. L. P. Santos, J. Wood, E. Selmanovic, C. Harvey, K. Debattista and A. Chalmers; **"Bespoke high-fidelity visualization of tiling"**; HDRi 2013: First International Conference and SME Workshop on HDR Imaging (short paper), **2013**

    https://core.ac.uk/download/pdf/55626497.pdf

    Scopus --- WEB OF SCIENCE --- 0

77. Maçães, Gustavo; Domingues, Hugo; Luis Almeida; Santos, Luís Paulo; **"COGNITO – Captura, reconhecimento e visualização de atividades manuais complexas"**; 20° Encontro Português de Computação Gráfica (short paper), **2012**

    https://diglib.eg.org/bitstream/handle/10.2312/pt20121115/115-118.pdf

    Scopus --- WEB OF SCIENCE --- 0

78. Santos, Luís Paulo and Coelho, Vítor and Bernardes, Paulo and Proença, Alberto; **"High Fidelity Walkthroughs in Archaeology Sites"**; VAST'2005 - 6th Int. Symposium on Virtual Reality, Archaeology and Cultural Heritage (short paper), **2005**

    https://core.ac.uk/download/pdf/55604667.pdf

    Scopus --- WEB OF SCIENCE --- 3

Posters

[Back to Publications]

79. Ramôa, Alexandra; Santos, Luís Paulo; Galvão, Ernesto; **"Bayesian Amplitude Estimation"**; EQTC' 2024 – European Quantum Technologies Conference (poster), **2024**

    EQTC' 2024

80. Coelho, Rodrigo; Santos, Luís Paulo; Sequeira, André; **"Tradeoff between moving targets, gradient magnitude and performance in quantum variational Q-Learning"**; QTML'2023 – 7th International Conference on Quantum Techinques in Machine Learning (poster), **2023**

    QTML' 2023

81. Sequeira, André; Santos, Luís Paulo; Barbosa, Luís Soares; **"Quantum Fisher kernels for improved classification and generalization"**; QTML'2023 – 7th International Conference on

Quantum Techinques in Machine Learning (poster), **2023**

QTML' 2023

82. Ramôa, Alexandra; Galvão, Ernesto; Santos, Luís Paulo; **"The impact of decoherence in quantum amplitude estimation algorithms"**; TQC 2023: the 18th Conference on Theory of Quantum Computation, Communication and Cryptography (poster), **2023**

    TQC 2023 Posters

83. Sequeira, André; Santos, Luís Paulo; Barbosa, Luís Soares; **"Barren plateaus in quantum policy gradients"**; TQC 2023: the 18th Conference on Theory of Quantum Computation, Communication and Cryptography (poster), **2023**

    TQC 2023 Posters

84. Coelho, Rodrigo; Sequeira, André; Santos, Luís Paulo; **"Data Re-Uploading in Quantum Variational Q-Learning"**; TQC 2023: the 18th Conference on Theory of Quantum Computation, Communication and Cryptography (poster), **2023**

    TQC 2023 Posters

85. Sequeira, André; Santos, Luís Paulo; Barbosa; Luís Soares **"Policy Gradients using Variational Quantum Circuits"**; QTML 2022: Quantum Techniques and Machine Learning (poster) – p. 10, **2022**

    PDF

86. Pimenta, Waldir; Santos, Luís Paulo; **"Evaluation of the compressibility of Computer-Generated Holograms"**; Eurographics 2015: the 36th Annual Conference of the European Association for Computer Graphics (poster), **2015**

    PDF

3.2 Community's Recognition

[Back to Scientific Results]

Description	Number
Awards	6
Associate Editor	1
Guest Editor	8
Scientific Commissions (chair)	8
Scientific Commissions (membro)	100
Projects' Evaluation	3
Invited Speaker (int.)	2
Invited Speaker (nat.)	10

Community's Recognition: fast access

Awards

[Back to Community's Recognition]

87. **Prémio "Melhor Artigo"**:

    Alves C., Santos L.P., Bashford-Rogers T.; **"A Quantum Algorithm for Ray Casting using an Orthographic Camera"** In: 2019 Int. Conf. on Graphics and Interaction (ICGI), **2019**

    doi 10.1109/ICGI47575.2019.8955061

88. **Prémio José Luís Encarnação 2014** : atribuído ao melhor artigo em co-autoria com um estudante na área da Computação Gráfica publicado em Portugal no período 2013-14, pelo Grupo Português de Computação Gráfica. 14, Novembro, 2014

    Marques, Ricardo; Bouville, Christian; Ribardiére, Mickael; Santos LP, Bouatouch, Kadi; **"A Spherical Gaussian Framework for Bayesian Monte Carlo Rendering of Glossy Surfaces"**, IEEE Transactions on Visualization and Computer Graphics, vol 19(10), pp. 1619-1632, October **2013**

    doi 10.1109/TVCG.2013.79

89. **Prémio José Luís Encarnação 2013** : atribuído ao melhor artigo em co-autoria com um estudante na área da Computação Gráfica publicado em Portugal no período 2012-13, pelo Grupo Português de Computação Gráfica. 7, Novembro, 2013

Nuno Silva, Luís Paulo Santos; **“Interactive High Fidelity Visualization of Complex Materials on the GPU”**; Elsevier Computers & Graphics, vol. 37(7), pp. 809-819, November **2013**

doi 10.1016/j.cag.2013.06.006

90. **Prémio ”Melhor Artigo”**:

    Silva, Nuno and Santos, Luís Paulo; **“Visualização em Tempo Real de um Modelo Esparso de Mistura Paramétrica para síntese da BTF”**; In: 20° Encontro Português de Computação Gráfica, **2012**

    doi 10.2312/pt.20121103

91. **Fraunhofer Portugal Challenge 2011** : 2° lugar na categoria Mestrado, pela orientação de Waldir Pimenta, com a dissertação intitulada:

    “Face detection, tracking and recognition through computer vision for video-surveillance applications”, outubro, **2011**

92. **Menção Honrosa** :

    Marques, Ricardo and Santos, Luís Paulo; **“GPU Ray Casting”**; 17° Encontro Português de Computação Gráfica, **2009**

    URL: doi 10.2312/pt.20091215

Associate Editor

[Back to Community’s Recognition]

93. Elsevier Computers & Graphics, Editor Associado; 2011 .. 2018

    http://www.journals.elsevier.com/computers-and-graphics/

    Publicação científica reconhecida pela comunidade científica, incluída em todos os rankings relevantes, tais como ISI Web of Knowledge, SCOPUS, SCIMAGO.

Guest Editor

[Back to Community’s Recognition]

94. Elsevier Computers & Graphics;

    Special Section on **Recent Advances in Graphics and Interaction** (Journal papers track for the International Conference on Graphics and Interaction 2021 – ICGI’2021); edited by Nuno Rodrigues, Daniel Mendes, Luís Paulo Santos and Kadi Bouatouch; vol. 102; October, **2021**

    doi 10.1016/j.cag.2021.12.006

95. Elsevier Computers & Graphics;

    Special Section on **The Spring Conference on Computer Graphics**; Eds.: Santos, Luís Paulo; Durikovic, Roman; vol. 53, June, **2015**

    doi 10.1016/j.cag.2015.05.030

96. Elsevier Computers & Graphics;

    High Dynamic Range Imaging; Eds.: Santos, Luís Paulo; Debattista, Kurt; vol. 37(7), November **2013**

    doi 10.1016/j.cag.2015.05.030

97. Elsevier Computers & Graphics;

    Special section on **Graphics for Cultural Heritage**; Eds.: Alan Chalmers, Mark Mudge, Luis Paulo Santos; Volume 36(1); February, **2012**

    doi 10.1016/j.cag.2011.11.006

98. Elsevier Computers & Graphics;

    Special section on **Cultural Heritage**; Eds.: Alan Chalmers, Mark Mudge, Luis Paulo Santos; Volume 35(4); August, **2011**

    doi /10.1016/j.cag.2011.03.041

99. Elsevier Computers & Graphics;

    Special section on **Graphics for Serious Games**; Eds.: Kurt Debattista, Alberto José Proença, Luis Paulo Santos; Volume 34(6) December, **2010**

    doi 10.1016/j.cag.2010.09.016

100. Elsevier Computers & Graphics;

    Special section on **Parallel Graphics and Visualization**; Eds.: Luis Paulo Santos, Dirk Reiners, Jean Favre; Volume 32(1), March, **2008**

    doi 10.1016/j.cag.2007.12.004

101. Journal of Parallel Computing;

    Special section on **Parallel Graphics and Visualization**; Eds.: Luis Paulo Santos, Alan Heirich, Bruno Raffin; Volume 33(6), June, **2007**

    doi 10.1016/j.parco.2007.04.001

Chair: scientific commissions

[Back to Community's Recognition]

102. Associate Area Chair for "Rendering"

 "GRAPP'25 – 20th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications", **2025**

103. Associate Area Chair for "Rendering"

 "GRAPP'24 – 19th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications", Rome, Italy, 27-29, Feb, **2024**

104. Copresidente da Comissão de Programa de artigos curtos

 "37th Annual Conference of the European Asscociation for Computer Graphics - EG'2016", Lisbon, Portugal, 9-13, May, **2016**

105. Copresidente da Comissão de Programa

 "31st Spring conference on Computer Graphics - SCCG'2015", Slomenice Castle, Slovakia, 22-24, April, **2015**

106. Copresidente da Comissão de Programa

 "Simpósio Ibero-Americano de Computação Gráfica - SIACG'2014", Bahia Blanca, Argentina, setembro, **2014**

107. Copresidente da Comissão de Programa

 "Eurographics Symposium on Parallel Graphics and Visualization - EGPGV'07", Lugano, Suíça, maio, **2007**

108. Copresidente da Comissão de Programa para artigos curtos

 Graphite'06, Kuala Lumpur, Malásia, dezembro, **2006**

109. Copresidente da Comissão de Programa

 Tópico "Scheduling and Load Balancing", "EuroPar'05", Lisboa, Portugal, setembro, **2005**

Member: scientific commissions

[Back to Community's Recognition]

110. SIBGRAPI – Conference on Graphics, Patterns, and Images;

 2018 ... 2024

111. GRAPP – International Conference on Computer Graphics Theory and Applications

2014 ... 2024

112. Short papers -– Annual Conference of the European Association for Computer Graphics

2017 ... 2022

113. International Conference on Graphics and Interaction (ICGI)

2018 ... 2024

114. Eurographics Workshop on Parallel Graphics and Visualization (EGPGV)

1996 ... 2023

115. IEEE Vis 2022: Visualization & Visual Analytics; Oklahoma City, USA;

2022

116. SeGaH –IEEE Int. Conf. on Serious Games and Applications for Health

2011 ... 2019

117. IEEE International Conference on Games and Virtual Worlds for Serious Applications (VS-GAMES)

2010 ... 2014, 2019

118. Encontro Português de Computação Gráfica

2003 .. 2017

119. 37th Annual Conference of the European Association for Computer Graphics - EG'2016, Lisbon, Portugal

2016

120. Sciences and Technologies of Interaction (SciTecIN'15)

2015

121. WSCG – International Conference on Computer Graphics, Visualization and Computer Vision

2012 ... 2014

122. International Conference and SME Workshop on HDR imaging (HDRi)

2013, 2014

123. VideoJogos

2013 ...2014

124. INForum **2013**

125. DSIE'2012 - Simpósio Doutoral em Engenharia Informática da FEUP
2012

126. "Simpósio Ibero-Americano de Computação Gráfica (SIACG)"
2009, 2011

127. Interação (Grupo Português de Computação Gráfica)
2008, 2010

128. Eurographics International Symposium on Virtual Reality, Archaeology and Cultural Heritage (VAST)
2008 ... 2011

129. Graphite, **2005 ... 2007**

130. Artech, **2006**

Projects' Evaluation

[Back to Community's Recognition]

131. Avaliador externo para o Programa UNA4CAREER, promovido pela Universidad Complutense de Madrid para atribuição de bolsas de pós-doutoramento aos seus Centros de Investigação avaliados com Excelente; Maio - **2021**

132. Membro do painel de avaliação da 7ª edição do Programa +Inovação +Indústria, promovido pela Portugal Ventures, na qualidade de perito na área da Computação Gráfica, junho, **2016**

133. Avaliador externo para a iniciativa DESMI 2009-2010, promovida pela Research Promotion Foundation, República do Chipre, cofinanciada pela República do Chipre e União Europeia; janeiro, **2010**

Invited Speaker (international)

[Back to Community's Recognition]

134. **"Quantum ray tracing"**; Invited presentation at Texas Advanced Computing Center (TACC), University of Texas at Austin (UTA), Austin, TX, USA;
22nd Apr.**2020**

135. **"Quantum computing fundamentals"**; Invited presentation at Texas Advanced Computing Center (TACC), University of Texas at Austin (UTA), Austin, TX, USA;
3rd Feb.**2020**

Invited Speaker (national)

[Back to Community's Recognition]

136. **"Quantum computing"**; Jornadas de Engenharia Física e Física (JEF^2)
Universidade do Minho, Portugal, 18 February, **2025**

137. **"Quantum computing: fundamentals and perspectives"**; MACC Users Group Workshop 2022
Porto, Portugal, 7 July, **2022**

138. **"Quantum Computing"**; Summer School on Machine Learning and Big Data with Quantum Computing – SMBQ'2020
Porto, Portugal, Setembro 7-8, **2020**

139. **"Tecnologia nos próximos 20 anos"**; Sociedade Civil, RTP 2, ep. 153
transmitido a 04.Nov.**2019**

140. **"Masterclass: Quantum Computing – principles, algorithms and applications"** ; UT Austin Portugal 2019 Annual Conference – Create Knowledge Foster Change
https://utaustinportugal.org/events/annu alconference2019/ Braga, Portugal, 20th.Sep.**2019**

141. **"Computação Quântica"**;
CTT - 3° Sessão de Inovação Exploratória
28 de junho, **2019**

142. **"Applied Visualization"**; UT Austin Portugal Applied Visualization Workshop
7-8.jun.**2019**

143. **"Grover's algorithm"**; QuantaLab Workshop in Quantum Computation – QDAYS'2019
Braga, Portugal, 11.Apr.**2019**

144. **"IBM Quantum Workshop @EDP"**; Formação para quadros superiores da EDP Inovação em Computação Quântica em co-coordenação com a IBM; IBM Portugal, Lisboa, Portugal
14..16.Janeiro, **2019**

145. **"Síntese de Imagens Fisicamente Corretas"**; Palestra convidada nas III Jornadas de Computação Gráfica e Multimédia, Instituto Politécnico de Viana do Castelo; abril, **2005**

3.3 Projects' Coordination and Participation

[Back to Scientific Results]

Description	Number
European Projects - participant	1
Advanced Computing national projects - PI	2
National R&D projects - PI	4
National R&D projects - participant	4
International Cooperation projects	4
INESC TEC *Seed Projects* - participant	1
Projects funded by the industry - participante	1
PhD grants - supervisor	7
Total :	**22**

Projects' coordination and participation

European Projects - participant

[Back to Projects' coordination and participation]

146. **"COGNITO - Cognitive Workflow Capturing and Rendering with On-Body Sensor Networks"**
TOTAL: 4.300.000,00 €(Universidade do Minho: 41.064 €)
FP7-ICT-2009-4, **2009..2012**

Advanced Computing national projects- principal investigator

[Back to Projects' coordination and participation]

147. **"Noise aware quantum algorithms and classical simulators in the NISQ era"**;
100.000 CPU.core hours on the RNCA HPC infrastructure; 2024.07005.CPCA, Jul.**2024**

148. **"Towards noise resilient quantum algorithms in the NISQ era"**;
100.000 CPU.core hours on the RNCA HPC infrastructure; 2023.09537.CPCA, Jul.**2023**

National R&D projects - principal investigator

[Back to Projects' coordination and participation]

149. **“PERFORM: Portability and Performance in Heterogeneous Many Core Systems”**;
Total: 100.000,00 €(Universidade do Minho: 75.688 €); PTDC/EIA-EIA/100035/2008, **2010..2012**

150. **“TopicShoe – Total Design, Process Integration & Commerce Support Tools for the Shoe Industry”**;
Total: 1.623.000,00 €(Universidade do Minho: 41.328,00 €) Projeto em Co-Promoção QREN – SI I&DT, **2010 .. 2013**

151. **”IGIDE: Interactive Global Illumination on Dynamic Environments”**;
Total: 101.000,00 €; PTDC/EIA/65965/2006; **2007..2010**

152. **”SIGMA - SIstema de Georeferenciação Móvel Assistido por imagem”**
Total: 165.268,37 €Agência de Inovação, POCTI 2.3 e POSI 1.3, **2003 .. 2005**

National R&D projects - participant

[Back to Projects’ coordination and participation]

153. **”Parallel Programming Refinements for Irregular Applications”**
95.000,00 €(Universidade do Minho: 76.104,00 €) ; UTAustin/CA/0056/2008; **2009..2012**

154. **”CROSSFIRE-Collaborative Resources Online to Support Simulations on Forest Fires: a Grid Platform to Integrate Geo-referenced Web Services for Real-Time Management”**
170.000,00 €; GRID/GRI/81795/2006; **2007..2009**

155. **“PPC-VM - Computação Paralela Portável Baseada em Máquinas Virtuais”**
73.690 €; POSI/CHS//47158/2002; **2004..2007**

156. **“ViAr - Affordable Interactive Virtual Archaeology with Adaptive Cluster Computing”**;
68.000 €; POSI/CHS/42041/2001; **2002..2004**

International cooperation projects

[Back to Projects’ coordination and participation]

157. **”PASP PALOP-TL – Projecto de Apoio à Melhoria da Qualidade e Proximidade dos Serviços Públicos dos PALOP-TL”**
Membro da equipa do projecto; **2016 .. 2017**

158. **"GCIO - Government Chief Information Officer"**;

Membro da equipa e responsável pela actividade "Comunidades de Prática"

Financiado pelo Programa das Nações Unidas para o Desenvolvimento (PNUD), com o objectivo de promover a excelência na eGov na Colômbia;

Orçamento UMinho e Unidade Operacional em Governação Electrónica (eGov) da Universidade das Nações Unidas: 202.278,00 €; GCIO.EDU.CO; **2015 .. 2016**

159. **"Síntese Interativa de Imagens com Meio Participativo"**

financiado no âmbito das Ações Integradas Luso-Britânicas 2009, "Tratado de Windsor", pelo CRUP e British Council; cooperação com Professor Alan Chalmers, The Digital Lab, University of Warwick, United Kingdom

Ação N° B-35/09, ; 1.500 €, **2009**

160. **"Síntese Interativa de Imagens de Alta Fidelidade"**;

financiado no âmbito das Ações Integradas Luso-Britânicas 2006, "Tratado de Windsor", pelo CRUP e British Council; cooperação com Professor Alan Chalmers, University of Bristol, United Kingdom

Ação No B-30/06, ; 1.450 €, **2006**

INESC TEC *Seed Projects* - participant

[Back to Projects' coordination and participation]

161. **"QuantELM: from Classic to Quantum Optical Extreme Learning Machines with integrated optics"**;

financiado pelo INESC TEC (*seed projects*); 36.000,00 euros; **2023**

Projects funded by the industry - participant

[Back to Projects' coordination and participation]

162. **"CAD2FE – inferring properties from CAD representations to Finite Element meshes"**;

financiado pela Ford Automotive, em colaboração com Instituto de Polímeros e Compósitos, Universidade do Minho; 9.000,00 euros; **2008 .. 09**

PhD grants - supervisor

[Back to Projects' coordination and participation]

163. Eduardo Araújo;

"Boundary between classical simulability and quantum computational advantage in near-term bosonic quantum computers"

Ref: 2025.07335.BDANA ; Bolsa de Doutoramento em Ambiente Não Académico: Instituto Internacional Ibérico de Nanotecnologia

Doutoramento em Informática; Universidade do Minho; **agosto, 2025 ... presente**

164. Diogo Manuel Pereira Gomes;

"Physics-inspired Quantum Algorithms for Near-term Quantum computers"

Ref: 2025.07323.BDANA ; Bolsa de Doutoramento em Ambiente Não Académico: Instituto Internacional Ibérico de Nanotecnologia

Doutoramento em Informática; Universidade do Minho; **agosto, 2025 ... presente**

165. Alexandra Francisco Ramôa da Costa Alves;

"Quantum algorithms for amplitude estimation in the NISQ context";

Ref: 2022.12332.BD ; Doutoramento em Informática; Universidade do Minho, desde setembro, **2022**

166. Mafalda Francisco Ramôa da Costa Alves;

"Quantum algorithms for optimization in the NISQ context";

Ref: 2022.12333.BD ; Doutoramento em Informática; Universidade do Minho, desde setembro, **2022**

167. André Manuel Resende Sequeira;

"Quantum Reinforcement Learning: Foundations, Algorithms, Applications";

Ref: UI/BD/152698/2022 ; Doutoramento em Informática; Universidade do Minho, desde **2022**

168. Perdigão, César;

"Sparse Reconstruction of Multidimensional Light Transport Signals";

ref: PD/BD/113966/2015 ; Doutoramento em Informática (MAPi); Universidade do Minho, **2017 ... 2020**

169. Waldir Spencer Pimenta Lima;

"Sparse Computer Generated Holography";

Ref: SFRH / BD / 74970 / 2010 ; Doutoramento em Informática (MAPi); Universidade do Minho, **2011 ... 2015**

3.4 Research Activities' Coordination

[Back to Scientific Results]

Description	Number
Organization of scientific events - chair	4
Organization of scientific events - member	5
Visiting researcher in international research centres	8
International scientific cooperation networks	3

Totals: Research Activities' Coordination

Organization of scientific events - chair

[Back to Research Activities' Coordination]

170. "Eurovis'2019: 21st EG/IEEE VGTC Conference on Visualization"

     Porto, Portugal, 3..7, june, **2019**

171. "International Conference on Theory and Practice of Electronic Governance (ICEGOV'2014)"

     Conferência co-organizada pela Universidade do Minho, Universidade das Nações Unidas e Agência para a Modernização Administrativa, com o Alto Patrocínio de sua Excelência o Presidente da República de Portugal Dr. Aníbal Cavaco Silva

     Guimarães, Portugal, 27..30, Outubro, **2014**

172. **"IEEE International Conference on Games and Virtual Worlds for Serious Applications (VS-Games'10)"**

     Braga, Portugal, março, **2010**

173. **"Eurographics Workshop on Parallel Graphics and Visualization 2006 (EGPGV'06)"**

     Braga, Portugal, **2006**

Organization of scientific events - member

[Back to Research Activities' Coordination]

174. CHI' 2024;

     Assistant to the General Chair;

     CHI is the major venue for Computer Human Interaction, sponsored by ACM SIGCHI and registering in excess of 5000 attendees. My role as assistant to the General Chair terminated on May 2023, as the General Chair had to be substituted.

Honolulu, Hawai'i, USA, **2024**

175. "IEEE VR 2021";

 Finance Chair; Lisbon, Portugal, 27.Mar .. 1.Apr, **2021**

176. "EuroGraphics 2016: The 37th Annual Conference of the European Association for Computer Graphics"

 Lisbon, Portugal, May **2016**

177. **"VAST 2008 - 9th Int. Symp. on Virtual Reality, Archaeology and Cultural Heritage"**;

 Braga, Portugal, 2..5 dezembro, **2008**

178. **"3rd IEEE International Conference on Application of Concurrency to System Design"**

 Guimarães, Portugal, 18 .. 20 junho, **2003**

Visiting researcher in international research centres

[Back to Research Activities' Coordination]

179. Texas Advanced Computing Center, University of Texas at Austin, TX, **USA**

 Invited Researcher; January .. April, **2020**

180. Texas Advanced Computing Center, University of Texas at Austin, TX, **USA**

 Invited Researcher; June .. July, **2017**

181. Université de Rennes I, Rennes, **França**

 Professor convidado (*maître de conférences*)

 a convite do Professor Kadi Bouatouch, 16 a 29 de junho, **2013**

182. Warwick Manufacturing Group, University of Warwick, **United Kingdom**

 Visiting fellow ; a convite do Professor Alan Chalmers; janeiro a maio, **2013**

183. Université de Rennes I, Rennes, **França**

 Professor convidado (*maître de conférences*)

 a convite do Professor Kadi Bouatouch; , junho, **2011**

184. Université de Rennes I, Rennes, **França**

Professor convidado (*maître de conférences*)

a convite do Professor Kadi Bouatouch; , junho..julho, **2010**

185. Université de Rennes I, Rennes, **França**

Professor convidado (*maître de conférences*)

a convite do Professor Kadi Bouatouch; , junho..julho, **2008**

186. Dept. of Computer Science, University of Bristol, **United Kingdom**

Visiting Fellow, parcialmente financiado pela Fundação para a Ciência e Tecnologia, bolsa SFRH/BPD/11622/2002

a convite do Professor Alan Chalmers ; março a julho, **2003**

International scientific cooperation networks

[Back to Research Activities' Coordination]

187. **IFIP Working Group on "Foundations of Quantum Computation"**

estabelecido no contexto dos Technical Committees TC1 (Foundations of Computer Science) e TC2 (Software: Theory and Practice), desde Março, **2022**

188. **Comissão Executiva do Eurographics** – The European Association for Computer Graphics

2016 … 2020

189. Membro do **Eurographics** – The European Association for Computer Graphics; desde **2006**

4 Pedagogical Activity

[Back to start]

This chapter characterizes the candidate's pedagogical activity, organized according to the following sections:

- Teaching activity;
- Teaching activity: assessment;
- Support materials;
- Pedagogical innovation and valorization;
- Coordination of pedagogic projects;
- MsC and PhD supervisions.
- Pedagogical Activity Awards.

4.1 Teaching Activity

[Back to Pedagogical Activity]

Teaching Activity

Higher Education 1st Cycle - Responsible by Course Unit

[Back to teaching activity]

190. **2021/22 .. presente** – Arquitetura de Computadores

     Licenciatura em Engenharia Informática, 2° ano, 1° semestre

191. **2011/12 .. 2020/21** – Arquitetura de Computadores

     Mestrado Integrado em Engenharia Informática, 2° ano, 1° semestre

192. **2008/09 .. 2010/11** – Arquitetura de Computadores

     Licenciatura em Engenharia Informática, 2° ano, 1° semestre

193. **2008/09** – Computação Gráfica

     Licenciatura em Engenharia Informática

     Licenciatura em Ciências de Computação

     3° ano, 2° semestre

194. **2001/02 .. 2003/04, 2006/07** – Arquitetura de Computadores

     Licenciatura em Engenharia de Sistemas e Informática, 2° ano, 1° semestre

195. **2004** – Algoritmos e Estruturas de Dados II

     Licenciatura em Engenharia Informática, 2° e 3° anos

     Universidade Nacional de Timor-Lorosae, Díli, Timor-Lorosae

     Ao abrigo do programa de cooperação da Fundação das Universidades Portuguesas (FUP) e a Universidade Nacional de Timor-Lorosae

Higher Education 1st Cycle - Responsible by Course Unit

[Back to teaching activity]

196. **2022/23 .. presente** – Engenharia Física Aplicada

     Mestrado em Engenharia Física, 2° ano, 1° semestre

197. **2021/22 .. presente** – Ciência de Dados Quântica

     Mestrado em Engenharia Física, 1° ano, 2° semestre

     Especialização em Física da Informação

198. **2021/22 .. presente** – Visualização e Iluminação

     Mestrado em Engenharia Informática, 1° ano, 2° semestre

     Perfil de Especialização em Computação Gráfica

199. **2012/13 .. 2020/21** – Visualização e Iluminação II

     Mestrado Integrado em Engenharia Informática, 4° ano, 2° semestre

     Perfil de Especialização em Computação Gráfica

200. **2007/08 .. 2011/12** – Iluminação e FotoRealismo

     Mestrado em Informática/Engenharia Informática

     Unidade Curricular de Especialização em Computação Gráfica

201. **2003/04 .. 2006/07** – Iluminação e FotoRealismo

     Mestrado em Computação Gráfica e Ambientes Virtuais

202. **2003/04 .. 2005/06** – Animação

     Mestrado em Computação Gráfica e Ambientes Virtuais

Higher Education 3rd Cycle - Responsible by Course Unit

[Back to teaching activity]

203. **2013/14** – Seminários

     Programa Doutoral em Informática – consórcio Minho-Aveiro-Porto (MAP-i)

204. **2010/11, 2013/14** – Tecnologias de Informação: Computação Gráfica

     Programa Doutoral em Informática – consórcio Minho-Aveiro-Porto (MAP-i)

Lecturing team member

[Back to teaching activity]

205. **2021/22 .. presente** – Visão por Computador e Processamento de Imagem

     Mestrado em Engenharia Informática, 1° ano, 2° semestre

     Perfil de Especialização em Computação Gráfica

206. **2018/19 .. 2020/21** – Computação Avançada

     Mestrado Integrado em Engenharia Informática, 4° ano, 1° semestre

     Perfil de Especialização em Ciências de Dados

207. **2016/17 .. 2018/19** – Tecnologias e Aplicações

     Mestrado Integrado em Engenharia Informática, 4° ano, 2° semestre

     Perfil de Especialização em Computação Gráfica

208. **2015/16 – 2020/21** – Sistemas de Computação

     Mestrado Integrado em Engenharia Informática, 1° ano, 2° semestre

209. **2011/12, 2014/15** – Sistemas de Computação

     Licenciatura em Engenharia Informática, 1° ano, 2° semestre

210. **2014/15** – Laboratório em Engenharia Informática

     Mestrado em Engenharia Informática, 2° ano, 1° semestre

211. **2010/11** – Laboratórios de Informática III

     Licenciatura em Engenharia Informática, 2° ano, 2° semestre

212. **2010/11** – Seminário

     Mestrado em Engenharia Informática, 2° ano, 1° semestre

     Mestrado em Informática, 2° ano, 1° semestre

4.2 Teaching activity: Assessment

[Back to Pedagogical Activity]

This section includes the candidate's individual assessment results, as reported by the University's Quality Assurance Internal System. These results are based on the assessment forms optionally filled in by the students at the end of each term.

This includes results since 2013/14 (the candidate was on sabbatical leave in 2012/13 and 2019/20). Individual results are presented for the pair Lecturer/Course Unit (CU), for each academic year. In particular, the presented results correspond to the students' assessment of the CU and the candidate.

The assessment forms were modified during this period. Up until 2017/18 the assessment scale included 6 levels, from 1 to 6, corresponding to "Completely Disagree" up to "Completely Agree". Since 2018/19 the scale includes 5 levels, from 1 to 5, corresponding to "Very Bad", "Bad", "Reasonable", "Good" and "Excellent". Two different tables are used below to accommodate these two different scales. Within these tables, the column labelled with "**#**" indicates the number of students that actually submitted their forms.

Besides the scales, the questions posed to the students also changed. Up to 2017/18 they read as:

Course Unit – Globally, I have a positive appreciation of this CU.

Lecturer – Globally, I have a positive appreciation of the lecturer's performance within this CU's context.

From 2018/19, these questions read as follows:

Course Unit – Pedagogically, this CU's functioning was adequate.

Lecturer – The lecturer **Luís Paulo Peixoto dos Santos** performed his responsibilities in an adequate manner.

The results of these pedagogic forms are covered by some degree of confidentiality. In the tables below the assessment reports from 2021/22 to 2023/24 are made available – hyperlink in the column "Degree". Upon demand by the jury, the candidate will supply the remaining reports. The results relative to the CU are included only for those CUs that are the candidate's responsibility.

All the results are very positive, with the average rating sitting, in most cases, between the two best levels of the respective Likert scale.

Acronym	Description
LEI	Licenciatura em Engenharia Informática
MiEI	Mestrado Integrado em Engenharia Informática
MEI	Mestrado em Engenharia Informática
MEFis	Mestrado em Engenharia Física
AC	Arquitectura de Computadores
CA	Computação Avançada
CDQ	Ciência de Dados Quântica
CG	Computação Gráfica
SC	Sistemas de Computação
TA	Tecnologias e Aplicações
VI	Visualização e Iluminação
VI II	Visualização e Iluminação II
VCPI	Visão por Computador e Processamento de Imagem

Table: acronyms

Year	Degree	CU	#	rating [1 .. 5] CU	Lecturer
2024/25	LEI	AC	142	4.3 / 5	4.4 / 5
	MEFis	CDQ		4.0 / 5	4.3 / 5
	MEI	VI	10	3.7 / 5	3.9 / 5
	MEI	VCPI	11	3.7 / 5	3.9 / 5
2023/24	LEI	AC	50	3.9 / 5	4.1 / 5
	MEFis	CDQ	5	4.2 / 5	4.4 / 5
	MEI	VI	18	4.1 / 5	4.4 / 5
	MEI	VCPI	19	4.2 / 5	4.5 / 5
2022/23	LEI	AC	132	4.3 / 5	4.6 / 5
	MEFis	CDQ	6	4.5 / 5	4.2 / 5
	MEI	VI	14	4.4 / 5	4.6 / 5
	MEI	VCPI	12	3.8 / 5	4.3 / 5
2021/22	MiEI	AC	41	4.6 / 5	4.7 / 5
	LEI	AC	153	4.2 / 5	4.6 / 5
	MEFis	CDQ	6	4.2 / 5	4.7 / 5
	MEI	VI	26	3.9 / 5	4.2 / 5
2020/21	MiEI	AC	195	4.1 / 5	4.4 / 5
	MiEI	SC	181	—	4.0 / 5
	MiEI	CA	11	—	4.3 / 5
	MiEI	VI II	7	3.7 / 5	4.1 / 5
	MEI	VI II	3	4.3 / 5	5.0 / 5
2018/19	MiEI	AC	111	4.3 / 5	4.7 / 5
	MiEI	SC	156	—	4.1 / 5
	MiEI	CA	14	—	4.6 / 5
	MiEI	VI II	6	3.2 / 5	3.5 / 5
	MiEI	TA	9	—	3.5 / 5

2018/19 .. 2024/25 ('#' - number of submitted forms)

				rating [1 .. 6]	
YUear	**Degree**	**CU**	**#**	**CU**	**Lecturer**
2017/18	MiEI	AC	84	4.7 / 6	5.3 / 6
	MiEI	SC	44	—	5.4 / 6
	MiEI	VI II	7	5.6 / 6	5.6 / 6
	MiEI	TA	5	—	5.6 / 6
2016/17	MiEI	AC	71	5.0 / 6	5.4 / 6
	MiEI	SC	46	—	5.1 / 6
	MiEI	VI II	6	5.3 / 6	5.3 / 6
	MiEI	TA	5	—	4.8 / 6
2015/16	MiEI	AC	40	5.3 / 6	5.7 / 6
	MiEI	SC	24	—	5.5 / 6
	MiEI	VI II	5	4.8 / 6	5.2 / 6
2014/15	LEI	AC	37	4.8 / 6	5.2 / 6
	LEI	SC	29	—	5.6 / 6
	MEI	VI II	2	6.0 / 6	6.0 / 6
2013/14	LEI	AC	9	4.6 / 6	5.8 / 6
	LEI	SC	35	—	5.5 / 6
	MEI	CG	3	5.7 / 6	6.0 / 6

2013/14 .. 2017/2018 ('#' - number of submitted forms)

4.3 Pedagogical Support Materials

[Back to Pedagogical Activity]

This section lists the most relevant pedagogical support materials produced within the context of Course Units under the responsibility of the candidate (see section secção 4.1).

Quantum Data Science
Computer Architecture
Visualization and Illumination

Pedagogical Support Materials

Quantum Data Science

[Back to support materials]

This Course Unit was designed and lectured for the first time in 2021/22. The following support material was developed:

213. Lectures: presentations;
214. Practical sessions: tutorials;
215. Project description.

Computer Architecture

[Back to support materials]

This course unit was redesigned by the candidate in 2011/12. The learning outcomes and syllabus were adapted to convey a high level conceptual vision: how do computer architecture features impact in the performance and coding of programs expressed on a high level programming language.

All support material was developed from scratch. The lessons learned through the years have been incrementally integrated onto this material. It is available at:

216. Lectures: presentations;
217. Practical sessions: tutorials.

Visualization and Illumination

[Back to support materials]

This Course Unit was initially prepared by the candidate to be taught in 2003/04, as part of the Master Degree in Computer Graphics and Virtual Environments. It went through several upgrades to conform to the goals of other degrees and to integrate advancements in the knowledge and practice of Global Illumination. Currently, it is offered as part of the major in Computer Graphics, Master in Informatics Engineering. The support materials are:

218. Lectures: presentations;
219. Practical sessions: tutorials;
220. Project description.

4.4 Pedagogical innovation and valorization

[Back to Pedagogical Activity]

Pedagogical training courses

[Back to pedagogical innovation and valorization]

221. **1 e 2 de junho, 2023** – Learning based on challenges (seminar)

     1ª edição, 7 horas

     Promovido pelo Centro de Inovação e Desenvolvimento do Ensino e Aprendizagem da Universidade do Minho (Centro IDEA-UMinho);

222. **19, outubro .. 25 de novembro, 2022** – Avaliação+ (Assessment+)

     1ª edição, 15 horas

     Promovido pelo Centro de Inovação e Desenvolvimento do Ensino e Aprendizagem da Universidade do Minho (Centro IDEA-UMinho);

223. **12 e 23 de julho de 2021** – Docência+ (Lecturing+)

     4ª edição, 20 horas

     Promovido pelo Centro de Inovação e Desenvolvimento do Ensino e Aprendizagem da Universidade do Minho (Centro IDEA-UMinho) e pelo Núcleo de Ensino e Aprendizagem do Gabinete do Reitor da Universidade de Aveiro.

     Certification Badge

224. **9 de julho de 2021** – *"Team-Based Learning"*

     4ª Edição das Jornadas Interinstitucionais de Desenvolvimento Pedagógico (3 horas)

225. **14 de julho de 2021** – *"Metodologias de ensino e de aprendizagem ativas: prbl, pbl e cbl"*

     4ª Edição das Jornadas Interinstitucionais de Desenvolvimento Pedagógico (3 horas)

226. **21.janeiro.2021** – *"Formação Docente Blackboard"*

     "À conversa com Teresa Monteiro (EENG) – realizando avaliações online";

     formação promovida pela Unidade de Serviço de Apoio às Atividades de Educação, Universidade do Minho

227. **9 .. 10.maio.2005** – *"Effective Teaching"*;

Richard Felder, North Carolina State University;

Rebecca Brent, Education Designs Inc., North Carolina;

International PostGraduate Program, Universidade do Minho

4.5 Coordination of pedagogic projects

[Back to Pedagogical Activity]

Coordination of pedagogic projects

Degrees Coordination

[Back to Coordination of pedagogic projects]

228. **2023/24 … now** – Licenciatura em Engenharia Física

     Director

229. **2023/24 … now** – Mestrado em Engenharia Física

     Director

230. **2021/22 .. 2022/23** – Licenciatura em Engenharia Física

     Member Directive Commission

231. **2021/22 .. 2022/23** – Mestrado em Engenharia Física

     Member Directive Commission

232. **2013 .. 2015** – Mestrado em Engenharia Informática

     Vice Director

233. **abril, 2011 .. Novembro, 2012** – Programa Doutoral em Informática

     Director

234. **Julho, 2010 .. Novembro, 2013** – Licenciatura em Engenharia Informática

     Vice Director

235. **2003..2007** – Mestrado em Computação Gráfica e Ambientes Virtuais

     Member Directive Commission

Creation and restructuring of degrees

[Back to Coordination of pedagogic projects]

236. **2025** – Licenciatura e Mestrado em Engenharia Física

     Responsible for the elaboration of the restructuring and reaccreditation proposal submitted to the Accreditation Agency (A3ES).

237. **2019** – Licenciatura e Mestrado em Engenharia Informática

     Member of the panel designated by the Department to create these 2 degrees.

238. **2009** – Programa Doutoral em Engenharia Informática

     Responsible for the elaboration of the proposal to create this doctoral programme, submitted to the Accreditation Agency (A3ES).

4.6 MsC and PhD supervisions

[Back to Pedagogical Activity]

Description	#
Current PhD students	5
Past PhD students	5
Current MsC students	8
Past MsC students	31

MsC and PhD supervisions

Current PhD students

[Back to supervisions]

239. **agosto, 2025 ...** – Eduardo Araújo
"Boundary between classical simulability and quantum computational advantage in near-term bosonic quantum computers"
Doutoramento em Informática; Universidade do Minho
Bolsa da Fundação para a Ciência e Tecnologia, Ref: 2025.07335.BDANA Bolsa de Doutoramento em Ambiente Não Académico: Instituto Internacional Ibérico de Nanotecnologia

240. **agosto, 2025 ...** – Diogo Manuel Pereira Gomes
"Physics-inspired Quantum Algorithms for Near-term Quantum computers"
Doutoramento em Informática; Universidade do Minho
Bolsa da Fundação para a Ciência e Tecnologia, Ref: 2025.07323.BDANA Bolsa de Doutoramento em Ambiente Não Académico: Instituto Internacional Ibérico de Nanotecnologia

241. **fevereiro, 2022 ...** – Alexandra Francisco Ramôa da Costa Alves
"Quantum algorithms for amplitude estimation in the NISQ context"
Doutoramento em Informática; Universidade do Minho
Bolsa da Fundação para a Ciência e Tecnologia, Ref: 2022.12332.BD

242. **fevereiro, 2022 ...** – Mafalda Francisco Ramôa da Costa Alves
"Quantum algorithms for optimization in the NISQ context"

Doutoramento em Informática; Universidade do Minho

Bolsa da Fundação para a Ciência e Tecnologia, Ref: 2022.12333.BD

243. **janeiro, 2011 ...** – João Barbosa

"Hardware Aware Scheduling in Heterogeneous Environments"

Doutoramento em Informática (MAPi); Universidade do Minho

Past PhD students

[Back to supervisions]

244. **janeiro, 2021 ... 13.maio, 2025** – Raman Choudhary

"Contextuality and Quantum advantage in the NISQ settings";

Doutoramento em Informática (MAPi); Universidade do Minho

Bolsa da Fundação para a Ciência e Tecnologia, *Quantum Portugal Initiative*

245. **janeiro, 2021 ... 24.março, 2025** – André Manuel Resende Sequeira

"Quantum Reinforcement Learning: Foundations, Algorithms, Applications"

Doutoramento em Informática; Universidade do Minho

Bolsa da Fundação para a Ciência e Tecnologia, Ref: UI/BD/152698/2022

246. **2020** – César Perdigão

"Sparse Reconstruction of Multidimensional Light Transport Signals"

Doutoramento em Informática (MAPi); Universidade do Minho

Bolsa da Fundação para a Ciência e Tecnologia, Ref: PD/BD/113966/2015

247. **2019** – Roberto Ribeiro

"Numerical Simulations on Heterogeneous Systems: dynamic workload and power management"

Doutoramento em Informática; Universidade do Minho

248. **2013** – Ricardo Marques

"Bayesian Monte Carlo Integration for Global Illumination"

Doutoramento em Informática; Université de Rennes I / INRIA Rennes

(em co-supervisão com Prof. Kadi Bouatouch)

Current MsC students

[Back to supervisions]

249. **2024/2025** – Artur Jorge Castro Leite

    "Ray Tracing and Global Illumination in ODIN"

    Mestrado em Engenharia Informática; Universidade do Minho

250. **2024/2025** – Mafalda Pinto Couto

    "Time of Arrival Measurements"

    Mestrado em Engenharia Física; Universidade do Minho

    (cosupervised by Lorenzo Maccone, Dipartimento di Fisica, Universita' di Pavia, Italia)

251. **2023/2024** – Diogo Pereira Matos

    "Extension of a Android camera application for ARML 2.0 support"

    Mestrado em Engenharia Informática; Universidade do Minho

252. **2023/2024** – Diogo Manuel Pereira Gomes

    "Fixed point quantum search"

    Mestrado em Engenharia Física; Universidade do Minho

253. **2023/2024** – Cristiano José Gonçalves Neiva Pereira

    "Interactive Physically-based Rendering on the Unreal Engine"

    Mestrado em Engenharia Informática; Universidade do Minho

254. **2023/2024** – Bernardo Manuel Barbosa Soares

    "Data Reuploading and Expressivity / Trainibility tradeoff on Policy Gradients"

    Mestrado em Engenharia Física; Universidade do Minho

255. **2023/2024** –Fernando Augusto Marques Lobo

    "Rendering skin on a pedagogical ray traced based renderer"

    Mestrado em Engenharia Informática; Universidade do Minho

256. **2023/2024** – Rosa Vitória Santos Sousa

    "Simulação de computação quântica com ruído usando somas de caminhos de Feynman"

    Mestrado em Engenharia Física; Universidade do Minho

Past MsC students

[Back to supervisions]

257. **2023** – Márcio Mano

    "ADAPT-VQC: Adaptive Variational Quantum Classifier"

    Mestrado Integrado em Engenharia Física; Universidade do Minho

258. **2023** – Ivo Alexandre Pereira Baixo

    "Vision System Hardware: Simulation and Test Automation"

    Mestrado em Engenharia Informática; Universidade do Minho

259. **2023** –Rodrigo da Silva Gomes Peres Coelho

    "Tradeoff between moving targets, gradient magnitude and performance in quantum variational Q-Learning"

    Mestrado Integrado em Engenharia Física; Universidade do Minho

260. **2023** – Inês Sofia Paiva Bastos

    "Visualização de Dados: Fatores Críticos de Sucesso"

    Mestrado em Engenharia de Sistemas; Universidade do Minho

261. **2023** –David Alves Campos Ferreira

    "Simulador de computador quântico de somas de caminhos de Feynman"

    Mestrado Integrado em Engenharia Física; Universidade do Minho

262. **2023** – Tomás Rodrigues Alves de Sousa

    "Classification and Clustering using Swap Test as distance metric"

    Mestrado Integrado em Engenharia Física; Universidade do Minho

263. **2022** – Renato Alberto Soares de Brito

    "Quantum Reinforcement Learning: a heuristic approach to solve deterministic MDPs"

    Mestrado Integrado em Engenharia Física; Universidade do Minho

264. **2021** – André Manuel Resende Sequeira

    "Quantum-enhanced Reinforcement Learning"

    Mestrado Integrado em Engenharia Física; Universidade do Minho

265. **2019** – Carolina Alves

"A Quantum Algorithm for Ray Casting using an Ortographic Camera"

Mestrado Integrado em Engenharia Física; Universidade do Minho

266. **2019** – Tiago Manuel da Silva Santos;

"Mobile Ray Tracing"

Mestrado em Engenharia Informática; Universidade do Minho

267. **2015** – Nuno Miguel Lima Cardoso

"Integração do OptiX no Maya: Síntese Interativa de Imagens de Alta Qualidade"

Mestrado em Engenharia Informática; Universidade do Minho

268. **2015** –Tiago Luís Oliveira Leite

"Laboratório Virtual de Holografia gerada por computador na Web";

Mestrado em Engenharia Informática; Universidade do Minho

269. **2015** – César Morais Perdigão;

"Algoritmos de Transporte de Luz Avançados em Plataformas Heterogéneas: Avaliação da Framework DICE"

Mestrado em Engenharia Informática; Universidade do Minho

270. **2014** – Mário João Guedes Pinto

"Aquisição Rápida 360° de Informação 3D usando uma Configuração Estática"

Mestrado em Engenharia Informática; Universidade do Minho

271. **2013** – Miguel Branco Palhas

"An Evaluation of the GAMA/StarPU Frameworks for Heterogeneous Platforms: the Progressive Photon Mapping Algorithm"

Mestrado em Engenharia Informática; Universidade do Minho

272. **2013** – Cruz, Eduardo José Tanque de Pádua

"Parallel Interactive Ray Tracing and Exploiting Spatial Coherence"

Mestrado em Engenharia Informática; Universidade do Minho

273. **2012** – Alves, Ricardo

"Memory Management and Caching for Heterogeneous Computing Systems"

Mestrado em Informática; Universidade do Minho

274. **2012** – Hélio Manuel Vilas

"Jogos de Computador para a Formação em Engenharia Informática"

Mestrado em Informática; Universidade do Minho

275. **2012** – Silva, Nuno

"Realistic and Interactive Visualization of Complex Materials"

Mestrado em Informática; Universidade do Minho

276. **2012** – Santos, Pedro

"Computer Generated Holography"

Mestrado em Informática; Universidade do Minho

277. **2011** – Ribeiro, Roberto

"Scheduling on Heterogeneous Parallel Systems"

Mestrado em Engenharia Informática; Universidade do Minho

278. **2010** – Ferreira, Vânio

"Mapping local thicknesses from CAD models to finite element meshes"

Mestrado em Engenharia Informática; financiado pela Ford Motors; Universidade do Minho

279. **2010** – Maçães, Gustavo

"Representação e Edição de Animações para sistemas de Realidade Virtual e/ou Aumentada"

Mestrado em Engenharia Informática, Universidade do Minho

financiado pelo projeto COGNITO – 7° Programa-Quadro União Europeia

280. **2010** – Pimenta, Waldir

"Face detection, tracking and recognition through computer vision for video-surveillance applications"

Mestrado em Engenharia Informática; Universidade do Minho

281. **2010** – Maia, Joaquim André

"Suppression of non relevant objects in a medical image reconstructed volumetric scene"

Mestrado em Engenharia Informática; Universidade do Minho

282. **2010** – Gomes, César

"Ray tracing Interativo Híbrido: GPU + CPU"

Mestrado em Engenharia Informática; Universidade do Minho

283. **2009** – Marques, Ricardo

“Interactive Volume Rendering for Medical Imaging”

Mestrado em Engenharia Informática; Universidade do Minho

284. **2009** – Nunes, Miguel

“Desenvolvimento e Avaliação do Kernel de um Ray Tracer Interativo”

Mestrado em Engenharia Informática; Universidade do Minho

285. **2006** – Oliveira, António

“Critérios de Refinamento Progressivo para Navegação Interativa em Modelos Virtuais Complexos”

Mestrado em Computação Gráfica e Ambientes Virtuais; Universidade do Minho

286. **2006** – Valentim, Sérgio

“Pré-Computação Paralela e Progressiva do Transporte da Radiância”

Mestrado em Computação Gráfica e Ambientes Virtuais; Universidade do Minho

287. **2005** – Coelho, Vítor Manuel Santos

“Navegação Interativa em Modelos Complexos”

Mestrado em Informática; Universidade do Minho

4.7 Pedagogical Activity Awards

[Back to Pedagogical Activity]

288. **2025** – Pedagogical Merit Award, Escola de Engenharia, Universidade do Minho, relative to academic year 2023/24

5 Other Activities

[Back to start]

- Training programs (as lecturer);
- Scientific dissemination;
- Academic evaluation activities;
- Academic staff evaluation activities ;
- Institutional Capacity Building;
- Higher Education Management activities;
- Scientific Organizations Management activities.

5.1 Training programs (as lecturer)

[Back to Other Activities]

289. **14..16.Janeiro, 2019** – IBM Quantum Workshop @EDP

     Formação para quadros superiores da EDP Inovação em Computação Quântica em co-coordenação com a IBM

     IBM Portugal, Lisboa, Portugal

5.2 Scientific dissemination

[Back to Other Activities]

290. **10 April 2025** – From Information Security to Quantum Adoption

     Mesa Redonda: Boosting the Adoption of Quantum Technologies

     Evento organizado pela Universidade Nova SBE, Fujitsu e MDS

     Nova SBE, Carcavelos, Portugal

291. **20 March, 2024** – O impacto da Computação Quântica

     Tertúlia - Semana da Ciência e Inovação

     Universidade do Minho, Portugal

292. **7 July, 2022** – Quantum computing: fundamentals and perspectives

     MACC Users Group Workshop 2022

     Porto, Portugal

293. **04.Nov.2019** – Tecnologia nos próximos 20 anos

     Sociedade Civil, RTP 2, ep. 153

294. **11.Apr.2019** – Grover's algorithm

     QuantaLab Workshop in Quantum Computation – QDAYS'2019

     Braga, Portugal

295. **28 de junho, 2019** – Computação Quântica

     CTT - 3º Sessão de Inovação Exploratória

5.3 Academic evaluation activities

[Back to Other Activities]

Description	#
PhD - Foreign institutions (main examiner)	5
PhD - Foreign institutions (jury member)	1
PhD - Portuguese institutions (main examiner)	4
PhD - Universidade do Minho (main examiner)	1
PhD - Portuguese institutions (jury member)	10
PhD - Universidade do Minho (jury member)	7
Masters foreign institutions (main examiner)	2
Masters portuguese institutions (main examiner)	8
Masters UMinho (main examiner)	4
Masters UMinho (chair)	62
Total :	**101**

Academic evaluation activities

PhD - Foreign institutions (main examiner)

[Back to Academic evaluation activities]

296. **2025** – Peng, Jingchao

 "Advancements in Infrared Image Processing with Deep Learning: Perception, Imaging and Fusion"

 Doctor of Philosophy in Engineering

 East China University of Science and Technology, Shanghai, China, and

 University of Warwick, Warwick, United Kingdom

297. **2025** – Napoli, Kevin

 "Towards the Efficient Adaptation of Offline Physically Based Methods for Real-time Rendering"

 Doctor of Philosophy

 Dep. of Computer Science, Faculty of ICT, University of Malta, Valleta, Malta

298. **2023** – Magro , Mark Charles

 "Real Time Global Illumination on Distributed Systems"

Doctor of Philosophy

Dep. of Computer Science, Faculty of ICT, University of Malta, Valleta, Malta

299. **2019** – Marnerides , Demetris

"Deep Learning for High Dynamic Range Imaging"

Doctor of Philosophy in Engineering

University of Warwick, Warwick, United Kingdom

300. **2016** – Saltimis, Pinar

"High Fidelity Sky Models"

Doctor of Philosophy in Engineering

University of Warwick, Warwick, United Kingdom

301. **2015** – Bugeja, Keith

"High Fidelity Graphics using Unconventional Distributed Rendering Approaches"

Doctor of Philosophy in Engineering

University of Warwick, Warwick, United Kingdom

PhD - Foreign institutions (jury member)

[Back to Academic evaluation activities]

302. **2013** – Marques, Ricardo

"Bayesian and Quasi Monte Carlo Spherical Integration for Global Illumination"

Thèse pour le grade de Docteur de l' Université de Rennes 1, Mention: Informatique

Université de Rennes 1, Rennes, France

PhD - Portuguese institutions (main examiner)

[Back to Academic evaluation activities]

303. **2025** – Cruz, Pedro

"Quantum Computation for Condensed Matter"

Tese de Doutoramento em Física

Faculdade de Ciências da Universidade do Porto

304. **2025** – Rahmani, Zeinab

"Secure Multiparty Computation Based on Quantum Technologies"

Tese de Doutoramento Engenharia Eletrotécnica; Universidade de Aveiro

305. **2016** – Barreira, João Carlos dos Santos

“Desenvolvimento de Aplicações de Realidade Aumentada Coerentes com o Contexto Ambiental”

Tese de Doutoramento em Informática; Universidade do Trás-Os-Montes e Alto Douro

306. **2015** – Dias, Sérgio Emanuel Duarte

”Geometric Representation and Detection Methods of Cavities on Protein Surface”

Universidade da Beira Interior, Covilhã, Portugal

307. **2015** – Costa, Vasco Alexandre da Silva

”Parallel Real Time Ray Tracing of Complex Scenes”

Instituto Superior Técnico, Lisboa, Portugal

PhD - Universidade do Minho (main examiner)

[Back to Academic evaluation activities]

308. **2007** – Valbom, Leonel

”Integração de Realidade Virtual no Desenvolvimento de um Modelo de Instrumento Musical Imersivo”

Escola de Engenharia, Universidade do Minho

PhD - Portuguese institutions (jury member)

[Back to Academic evaluation activities]

309. **2024** – Peres, Filipa Cavaco Reis

“Optimizing Models of Hybrid Quantum / Classical Computation”

Tese de Doutoramento em Física

Faculdade de Ciências da Universidade do Porto

310. **2016** – Carvalho, Diana Carneiro Machado de

“Proposal of a Novel Multimodal InteractionModel for Users of Different Age Groups”

Tese de Doutoramento em Informática

Universidade do Trás-Os-Montes e Alto Douro

311. **2016** – Arabnejad, Hamid

“QoS based workflow scheduling on heterogeneous resources”

Tese de Doutoramento MAP-i Doctoral Program in Computer Science

Faculdade de Engenharia da Universidade do Porto

312. **2015** – Adão, Telmo Miguel Oliveira

“Ontology-based Procedural Modelling of Traversable Buildings Composed by Arbitrary Shapes”

Tese de Doutoramento em Informática

Universidade do Trás-Os-Montes e Alto Douro

313. **2015** – Gonçalves, Martinho Fradeira

“Reorganization of Web sites content based on Users Visual Behaviour”

Tese de Doutoramento em Informática;

Universidade do Trás-Os-Montes e Alto Douro

314. **2014** – Rocha, Tânia de Jesus Vilela da

“Metáfora de Interacção para o Acesso à informação Digital de uma Forma Autónoma por Pessoas com Deficiência Intelectual”

Tese de Doutoramento em Informática; Universidade do Trás-Os-Montes e Alto Douro

315. **2013** – Urbano, António Carlos Alves

“Visualização de Imagens HDR em Dispositivos com Ecrã Pequeno”

Tese de Doutoramento em Informática; Universidade do Trás-Os-Montes e Alto Douro

316. **2010** – Gonçalves, Alexandrino José Marques

”Perceptually Valid Images of Conimbriga using High Dynamic Range Images”

Universidade do Trás-Os-Montes e Alto Douro

317. **2007** – Bessa, Maximino Esteves Correia

”Selective Rendering for High-Fidelity Graphics for 3D Maps on Mobile Devices”

Universidade do Trás-Os-Montes e Alto Douro

PhD - Universidade do Minho (jury member)

[Back to Academic evaluation activities]

318. **2025** – Wagner, Rafael

“Coherence and contextuality as quantum resources”

Programa Doutoral Minho-Aveiro-Porto em Física (MAPfis); Universidade do Minho

319. **2025** – Guimarães, José Diogo da Costa

“Simulation of open systems using near-term quantum computers with applications to the energy transport in photosynthetic systems"

Programa Doutoral Minho-Aveiro-Porto em Física (MAPfis); Universidade do Minho

320. **2022** – Silva, Rui António Sabino Castiço

"Locality optimisation techniques for platforms"

Programa Doutoral em Informática; Universidade do Minho

321. **2019** – Pereira, André Martins

"HEP-Frame: a Development Aid and Efficient Execution Engine where a Multi Layer Scheduler Adaptively Orders a Pipelined Data Stream Application"

Programa Doutoral Minho-Aveiro-Porto em informática (MApi); Universidade do Minho

322. **2018** – Tavares, Paula Correia

"O Impacto da Animação e da Avaliação Automática na Motivação para o Ensino da Programação"

Doutoramento em informática; Universidade do Minho

323. **2009** – Deusdado, Leonel Domingues

"Ambientes Virtuais Povoados com Simulação Eficiente de Deteção de Colisões e Planeamento de Trajetos em Navegação Realmente 3D"

Universidade do Minho

324. **2008** – Rufino, José

"Co-Operação de Tabelas de Hash Distribuídas em Clusters Heterogéneos"

Universidade do Minho

325. **2006** – Rodrigues, Rui Pedro Amaral

"Robust and hardware accelerated 3D point and line reconstruction from images"

Universidade do Minho

Masters foreign institutions (main examiner)

[Back to Academic evaluation activities]

326. **2013** – Daniel D'Agostino

"Distributed High-Fidelity Graphics using P2P"

University of Malta; Faculty of ICT; Department of Computer Science

327. **2013** – Kovanda, Jan

"Optimization of Light Propagation Computation"

University of West Bohemia; Faculty of Applied Sciences; Department of Computer Science and Engineering

Masters portuguese institutions (main examiner)

[Back to Academic evaluation activities]

328. **2023** – Luis Miguel Bastos Barrancos Fernandes

"Enhancing Mesh Deformation Realism: Dynamic Mesostructure Detailing and Procedural Microstructure Synthesis"

Mestrado em Multimédia - Especialização em Tecnologias Interativas e Jogos Digitais; Universidade do Porto

329. **2023** – Tomás Costa Fontes

"Augmented Reality Solution for the Côa Valley Rock Art"

Mestrado em Engenharia Informática e Computação; Faculdade de Engenharia, Universidade do Porto

330. **2020** – Vanda Filipa Bernardo Azevedo

"Quantum Transfer Learning for Breast Cancer Detection"

Mestrado em Ciências de Computação; Faculdade de Ciências, Universidade do Porto

331. **2013** – Nuno Filipe Frutuoso Ribeiro

"Visualização de dados sensoriais em aplicações de Realidade Aumentada"

Mestrado em Comunicação e Multimédia; Universidade de Trás-os-Montes e Alto Douro

332. **2013** – Marta Daniela Mendes Noval

"Realidade Aumentada no ensino da Matemática: um caso de estudo"

Mestrado em Tecnologias da Informação e Comunicação; Universidade de Trás-os-Montes e Alto Douro

333. **2013** – Pedro João Nunes Tavares

"Triangulação de Superfícies Implícitas com Singularidades"

Mestrado em Engenharia Informática; Universidade da Beira Interior

334. **2013** – Estevinho, João Paulo Abreu Nogueiro

“Utilização de Computação Heterogénea na Codificação de Vídeo”

Mestrado integrado em Engenharia Informática e Computação (MIEIC), FEUP

335. **2013** – Pires, António dos Santos Monteiro Borges

“Improving Electrical Production forecasting using GPU”

Mestrado integrado em Engenharia Informática e Computação (MIEIC), FEUP

Masters UMinho (main examiner)

[Back to Academic evaluation activities]

336. **2021** – José Eduardo Pereira

“Quantum Error-Correcting Codes”

Mestrado em Engenharia Física; Universidade do Minho

337. **2014** – Diogo Alberto Rocha Lopes

“Estado da arte em Dispersão Atmosférica”

Mestrado em Engenharia Informática; Universidade do Minho

338. **2014** – André Alexandre Wang Liu

“Debugging de Aplicações 3D”

Mestrado em Engenharia Informática; Universidade do Minho

339. **2009** – Ferreira, Sérgio Vasco Freitas

”Simulating High Quality Real-Time Shader Materials”

Mestrado em Computação Gráfica e Ambientes Virtuais, Universidade do Minho

Masters UMinho (chair)

[Back to Academic evaluation activities]

340. **2025** – Salgado, Bruna Filipa Martins

“Metric λ-calculus with conditionals: quantum, probabilities and beyond”

Mestrado em Engenharia Física, Universidade do Minho

341. **2025** – Magalhães, Ricardo da Silva

“Semi-transparent Cu(In,Ga)Se2 Solar Cells for agricultural-integrated photovoltaics applications”

Mestrado em Engenharia Física, Universidade do Minho

342. **2025** – Pereira, Anabela Sampaio

"Development of a hydrophone for measuring the propagation of acoustic waves in biological tissues"

Mestrado em Engenharia Física, Universidade do Minho

343. **2025** – Araújo, André Semanas de Oliveira

"DrumLace: a System for Drum Programming"

Mestrado em Engenharia Física, Universidade do Minho

344. **2025** – Silva, André Sá

"Fine-tunning of a selenization process to develop Cu(In,Ga)Se2 thin film solar cells"

Mestrado Integrado em Engenharia Física, Universidade do Minho

345. **2025** – Cunha, Orlando Daniel Sousa da

"Pulsed optically detected magnetic resonance (ODMR) protocols and experiments for advanced magnetic field characterization using Nitrogen-vacancies in diamond for the characterization of smart magnetic materials"

Mestrado em Engenharia Física, Universidade do Minho

346. **2025** – Portela, Maria João Barroso

"Verified Quantum Computing"

Mestrado em Engenharia Física, Universidade do Minho

347. **2025** – Carvalho, Nuno Filipe Quintela Teixeira

"Performance and Carbon Footprint Modelling Irregular Tasks"

Mestrado em Engenharia Física, Universidade do Minho

348. **2024** – Silva, Guilherme de Bastos

"Inspeção remota de processo de coating numa linha de produção usando técnicas de análise de imagem"

Mestrado em Engenharia Física, Universidade do Minho

349. **2024** – Martins, Hélder Vieira de Freitas Maia

"Quantum Resilient Security Proofs"

Mestrado em Engenharia Física, Universidade do Minho

350. **2024** – Moura, Ana Margarida Santos de

"Growth and characterization of a low-temperature process for Cu(In,Ga)Se2 thin film solar cells by pulsed hybrid magnetron sputtering"

Mestrado em Engenharia Física, Universidade do Minho

351. **2024** – Peixoto, Miguel Caçador

"Anomaly Detection as a Quality Control Tool in an Industrial Context"

Mestrado em Engenharia Física, Universidade do Minho

352. **2024** – Araújo, Eduardo Vieira

"Advantages and Limitations of Spatial Search by Continuous-Time Quantum Walk"

Mestrado em Engenharia Física, Universidade do Minho

353. **2024** – Oliveira, Maria Gabriela Jordão

"Quantum computing applications to quantum chromodynamics: Jet evolution in a quantum computer"

Mestrado em Engenharia Física, Universidade do Minho

354. **2024** – Dias, Maria Inês Machado Correia Brioso

"An Interpreter for a Concurrent Quantum Language"

Mestrado em Engenharia Física, Universidade do Minho

355. **2024** – Barros, João Carlos Macedo

"Magnetoliposomes as New Approach for Bone Cancer Therapies"

Mestrado em Engenharia Física, Universidade do Minho

356. **2023** – Valencia, Irving Leander Reascos

"Quantum Simulation of spin systems on Quantum Computers"

Mestrado em Engenharia Física, Universidade do Minho

357. **2023** – Correia, Ricardo da Silva

"Simulation of Hybrid Systems Regulated by Newtonian Mechanics"

Mestrado em Engenharia Física, Universidade do Minho

358. **2023** – Sampaio, Pedro António Ribeiro da Costa

"Desenvolvimento de baterias flexíveis sustentáveis para wearables"

Mestrado em Engenharia Física, Universidade do Minho

359. **2023** – Rodrigues, Nuno Oliveira

“Semi-transparent Cu(In,Ga)Se2 Solar Cells for Window Applications”

Mestrado em Engenharia Física, Universidade do Minho

360. **2023** – Ribeiro, Sara Coelho

“FAbrication of PEDOT:PSS/silver nanowire based films for the development of transparent heating systems”

Mestrado em Engenharia Física, Universidade do Minho

361. **2023** – Duarte, Magda Mendes

“Muon tomography: application of image reconstruction algorithms within the LouMu project”

Mestrado em Engenharia Física, Universidade do Minho

362. **2023** – Gil, Carlos Jorge Dias

“Deep Sea Biological Sampling”

Mestrado Integrado em Engenharia Física, Universidade do Minho

363. **2023** – Ribeiro, Clarisse Rodrigues

“Sensor de microplásticos para oceanos”

Mestrado em Engenharia Física, Universidade do Minho

364. **2023** – Carneiro, Tomás Barros

“iQbricks: Integration of a fully-featured quantum language in the framework Qbricks”

Mestrado Integrado em Engenharia Física, Universidade do Minho

365. **2022** – Cunha, Gilberto Rui Nogueira

“Quantum Bayesian Reinforcement Learning”

Mestrado Integrado em Engenharia Física, Universidade do Minho

366. **2019** – Cancelinha, Bruno Miguel Sousa

“Towards Model Checking Electrum Specifications with LTSmin”

Mestrado Integrado em Engenharia Informática, Universidade do Minho

367. **2019** – Maia, Octávio José Azevedo

“CLAV: Autenticação e integração na plataforma iAP”

Mestrado Integrado em Engenharia Informática, Universidade do Minho

368. **2019** – Cunha, José Manuel Gonçalves Leitão

"Discalculia em Crianças -Aplicativo móvel baseado em jogos para aprender matemática"

Mestrado Integrado em Engenharia Informática, Universidade do Minho

369. **2019** – Silva, Luís Martinho de Aragão Rego

"Intelligent feedback system for programmer's profile improvement"

Mestrado Integrado em Engenharia Informática, Universidade do Minho

370. **2015** – Santos, André Filie Faria

"Aplicação de Convenções de Código ao Robot Operating System"

Mestrado em Engenharia Informática, Universidade do Minho

371. **2015** – Miranda, João Pedro Lopes

"Replicação de base de dados baseada em comunicação em grupo"

Mestrado em Engenharia Informática, Universidade do Minho

372. **2015** – Neves, Francisco Nuno Teixeira

"Análise de desempenho e otimização do Apache HBase para dados relacionais"

Mestrado em Engenharia Informática, Universidade do Minho

373. **2014** – Amorim , Nuno Tiago Mourão de

"Georreferenciaçção por Imagem"

Mestrado em Engenharia Informática, Universidade do Minho

374. **2014** – Sousa, Nuno Miguel Rocha de

"Bidirectional Distributed Data Aggregation"

Mestrado em Engenharia Informática, Universidade do Minho

375. **2019** – Murta, Daniel Rodrigues Pacheco

"A comparison between DSLs and GPLs for the implementation of unidirectional and bidirectional transformations"

Mestrado em Engenharia Informática, Universidade do Minho

376. **2014** – Jorge, Tiago Manuel da Silva

"A Model Repair Application Scenario with PROVA"

Mestrado em Engenharia Informática, Universidade do Minho

377. **2014** – Cerqueira , Fábio Manuel Afonso

"Caching em Redes Tolerantes a Atrasos com Dados Nomeados"

Mestrado em Engenharia Informática, Universidade do Minho

378. **2014** – Loureiro, Flávio Roberto Ribeiro

"Utilização de Técnicas de Mineração Incremental em Aplicações de Previsão de Consumo de Energia Elétrica"

Mestrado em Engenharia Informática, Universidade do Minho

379. **2014** – Carvalho, Mickael Alexandre Silva

"Previsão em Tempo Real da Qualidade de Efluentes de uma ETAR"

Mestrado em Engenharia Informática, Universidade do Minho

380. **2014** – Fernandes, Bruno Filipe Martins

"Tratar informação desconhecida através de Raciocínio Baseado em Casos: uma abordagem para resolver Problem Reports"

Mestrado em Engenharia Informática, Universidade do Minho

381. **2014** – Campos, Tiago António Sousa

"Robô Interativo"

Mestrado em Engenharia Informática, Universidade do Minho

382. **2014** – Gonçalves, Manuel Francisco Mendes

"HTML5 - Análise dos Riscos de Segurança & Testes de Penetração a Aplicações Web"

Mestrado em Engenharia Informática, Universidade do Minho

383. **2014** – Albuquerque, Diego de Lara e

"Automatic detection of architectural violations in evolutionary systems"

Mestrado em Engenharia Informática, Universidade do Minho

384. **2014** – Pereira, Luís Tiago Gonçalves

"FITTEST logging and log-analysis infrastructure for PHP"

Mestrado em Engenharia Informática, Universidade do Minho

385. **2014** – Gomes, Tiago Emanuel Oliveira

"Geração de Ambientes Virtuais 3D"

Mestrado em Engenharia Informática, Universidade do Minho

386. **2013** – Silva, Nuno Miguel Milhases da

"Suporte à Interoperabilidade entre o Automation Studio e Sistemas SCADA: Tradução e sinópticos de XAML para SVG"

Mestrado em Engenharia Informática, Universidade do Minho

387. **2013** – Cruz, Paulo Filipe de Jesus

"Desenvolvimento de um Ambiente para a Geração, Mutação e Execução de Casos de Testes"

Mestrado em Engenharia Informática, Universidade do Minho

388. **2013** – Sampaio, Ana Isabel Anjos

"Responsive Web Design"

Mestrado em Engenharia Informática, Universidade do Minho

389. **2013** – Amaral, Vasco Manuel de Frias

"Framework e Cliente WebRTC"

Mestrado em Engenharia Informática, Universidade do Minho

390. **2013** – Silva, Tiago Fontes Carvalho Duque da

"Gestão Remota para Pontos de Acesso de Redes sem Fios"

Mestrado em Engenharia Informática, Universidade do Minho

391. **2013** – Dias, Miguel Gonçalves

"Captura de Dados em Tempo Real em Sistemas de Data Warehousing"

Mestrado em Engenharia Informática, Universidade do Minho

392. **2013** – Morgado, André Correia

"Data Warehouse Espacial"

Mestrado em Engenharia Informática, Universidade do Minho

393. **2013** – Liu, Filipe Alexandre Wang

"Computacional tools for pathway optimization towards metabolic engineering applications"

Mestrado em Engenharia Informática, Universidade do Minho

394. **2013** – Almeida, Paulo Adelino Dias de

"Um Globo Virtual Open Source para Android"

Mestrado em Engenharia Informática, Universidade do Minho

395. **2013** – Caneiro, Marco Pereira

"Um player web para redes de ecrãs públicos"

Mestrado em Engenharia Informática, Universidade do Minho

396. **2013** – Felgueiras, Gilberto Martins

"Social Simulation of human behavior in virtual agents for sustainability management platforms"

Mestrado em Engenharia Informática, Universidade do Minho

397. **2013** – Rosa, Luís Miguel Ferreira

"Sensorização, fusão sensorial e dispositivos móveis: contribuições para a sustentabilidade de ambientes inteligentes"

Mestrado em Engenharia Informática, Universidade do Minho

398. **2013** – Boas , Tiago Campelo Vilas

"Development and Implementation of Automatized Clinical Practice Guidelines"

Mestrado em Engenharia Informática, Universidade do Minho

399. **2009** – Pereira, Marta Susana Gonçalves

"Motor de Eventos"

Mestrado em Computação Gráfica e Ambientes Virtuais, Universidade do Minho

400. **2009** – Oliveira, Bruno Miguel Ferreira

"The anatomy of a real-time rendering engine"

Mestrado em Computação Gráfica e Ambientes Virtuais, Universidade do Minho

401. **2009** – Silva, Marco

"Domeview: a Digital Planetarium"

Mestrado em Computação Gráfica e Ambientes Virtuais, Universidade do Minho

402. **2009** – Brito, José Henrique Araújo Silveira de

"Development of a Prototype for Human Motion Capture using Computer Vision"

Mestrado em Computação Gráfica e Ambientes Virtuais, Universidade do Minho

5.4 Academic staff evaluation activities

[Back to Other Activities]

403. 2025 - Aviso n.º AE2025/0270 - Contratação de 1 Investigador(a) Auxiliar na Área de Computação Quântica, posição 2023.14760.TENURE.010, INESC TEC.

404. 2025 - Aviso n.º 6391/2025/2 - concurso para investigador auxiliar, na área científica de Tecnologias Digitais, subárea de Multimédia e Computação Gráfica, Universidade de Trás-os-Montes e Alto Douro.

405. 2025 - Edital n.º 1909/2024 - Concurso documental, de âmbito internacional, de recrutamento para um posto de trabalho de Professor Auxiliar na área disciplinar de Informática da Escola de Engenharia, Universidade do Minho.

5.5 Institutional Capacity Building

[Back to Other Activities]

406. **2016** – University of Minho Special Project for Electronic Governance (UMinho-EGOV)
Membro da Direcção (Despacho Reitoral 36/2016)
Este projecto especial, que funcionou na dependência direta do Reitor, corresponde ao envolvimento do candidato numa iniciativa de **capacitação institucional**, com o objectivo de instalar num campus da Universidade do Minho de uma Unidade Especial em Governação Electrónica da Universidade das Nações Unidas. A Unidade UNU–EGOV foi criada em 2014, no Campus de Couros, Guimarães, mantendo-se em operação até esta data.

5.6 Higher Education Management activities

[Back to Other Activities]

407. **Out.2010 — Nov.2012** – Vice Director: Departamento de Informática
Escola de Engenharia, Universidade do Minho

5.7 Scientific Organizations Management activities

[Back to Other Activities]

408. **2017 — 2020** – Eurographics – The European Association for Computer Graphics
Membro da Comissão Executiva

409. **biénios 2017/18 e 2019/20** – Grupo de Português de Computação Gráfica
Presidente eleito

410. **biénio 2015/16** – Grupo de Português de Computação Gráfica
Vice-Presidente eleito

411. **biénio2 2009/10, 2011/12 e 2013/14** – Grupo de Português de Computação Gráfica
Membro eleito da Direção – Tesoureiro

412. **biénio 2005/07** – Centro de Computação Gráfica
Vogal do Conselho de Gestão de Projetos

Luís Paulo Peixoto dos Santos

January 2026